

Large Language Models for Market Research: A Data-Augmentation Approach

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Received: January 6, 2025

Revised: October 19, 2025; January 6, 2026

Accepted: January 8, 2026

Published Online in *Articles in Advance*:
March 17, 2026

<https://doi.org/10.1287/mksc.2025.0009>

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Abstract. Large language models (LLMs) have transformed artificial intelligence by excelling in complex natural language processing tasks. Their ability to generate human-like text has opened new possibilities for market research, particularly in conjoint analysis, where understanding consumer preferences is essential but often resource intensive. Traditional survey-based methods face limitations in scalability and cost, making LLM-generated data a promising alternative. However, although LLMs have the potential to simulate real consumer behavior, recent studies highlight a significant gap between LLM-generated and human data, with biases introduced when substituting between the two. In this paper, we address this gap by proposing a novel statistical data augmentation approach that efficiently integrates LLM-generated data with real data in conjoint analysis. This results in statistically robust estimators with consistent and asymptotically normal properties, in contrast to naïve approaches that simply substitute human data with LLM-generated data, which can exacerbate bias. We further present a finite-sample performance bound on the estimation error. We validate our framework through an empirical study on COVID-19 vaccine preferences, demonstrating its superior ability to reduce estimation error and save data and costs by 24.9%–79.8%. In contrast, naïve approaches fail to save data due to the inherent biases in LLM-generated data compared with human data. Another empirical study on sports car choices validates the robustness of our results. Our findings suggest that, although LLM-generated data are not a direct substitute for human responses, they can serve as a valuable complement when used within a robust statistical framework.

History: Hema Yoganarasimhan served as the senior editor.

Funding: This work was supported by OpenAI Research funding.

Supplemental Material: The online appendices and data files are available at <https://doi.org/10.1287/mksc.2025.0009>.

Keywords: conjoint analysis • data augmentation • large language models

1. Introduction

Large language models (LLMs) have revolutionized artificial intelligence (AI) by delivering unprecedented capabilities in natural language processing (NLP). These models are built on advanced deep learning architectures known as transformer networks, which excel at handling sequential data and understanding context (Vaswani et al. 2017, Radford et al. 2018). Trained on extensive and diverse data sets—including vast amounts of text from books, articles, and websites—LLMs are capable of generating human-like responses and performing complex language tasks. For example, Llama 3 was trained on 15 trillion tokens of data (HuggingFace 2024). This training enables LLMs to comprehend and produce coherent, contextually relevant text, making them invaluable tools in a wide range of applications (Brown et al. 2020).

The rapid development of LLMs also signifies a transformative moment in the social sciences. In market research—a critical area in marketing science and operations management—understanding market demand for products or services is paramount. Traditional market research methods, such as surveys, focus groups, and conjoint analysis, often require significant resources and time. Gathering data from real subjects involves designing intricate questionnaires, recruiting participants, and analyzing responses, all of which can be both costly and logistically challenging (Hair et al. 2019). LLMs present a novel solution by enabling the generation of synthetic data that simulates real consumer behavior without the need for extensive fieldwork. This potential has sparked growing interest among researchers. Thus, a key advantage of using LLMs in market research is their ability to

scale data generation: Whereas traditional methods often limit the volume of data that can be feasibly collected due to constraints on time and resources, LLMs can rapidly generate large volumes of data, facilitating more comprehensive and granular analysis.

LLMs are trained on vast amounts of data from diverse sources across the internet, including reviews, purchase decisions, and even real survey data on products and services from various categories. The underlying assumption, expressed with some exaggeration, is that LLMs have been exposed to everything there is to see. Researchers hypothesize that the responses LLMs generate for market research surveys could, to some extent, reflect the types of responses that real consumers—represented in the training data—would have given to the same questions. These factors suggest that LLMs may become invaluable tools for gaining insights into consumer preferences, thanks to their ability to mimic or replicate human responses (Brand et al. 2023). Recent studies provided encouraging evidence supporting this potential (Argyle et al. 2023, Chen et al. 2023, Horton 2023). For instance, research showed that by fine-tuning a model (Gururangan et al. 2020) or employing prompting techniques such as chain-of-thought (CoT) reasoning (Wei et al. 2022), it is possible to generate simulated responses in market surveys that resemble those of real human subjects (Brand et al. 2023, Goli and Singh 2024). These explorations suggest the enormous potential of LLMs for advancing market research.

Nonetheless, recent literature also highlights several caveats. Despite some encouraging findings by Goli and Singh (2024, p. 1), the authors acknowledged that even the most up-to-date prompt engineering methods and advanced LLMs, such as OpenAI's Generative Pre-trained Transformer (GPT), cannot completely eliminate discrepancies between human and LLM responses in market surveys. For example, they observe that GPT-4 exhibits a pronounced level of impatience compared with human participants. This led the authors to conclude that “directly eliciting preferences using LLMs can yield misleading results,” suggesting that LLMs should be used more as tools to facilitate hypothesis exploration rather than as direct substitutes for human responses. Similarly, Gui and Toubia (2023) pointed out that LLMs tend to make unintended assumptions about unspecified details of the choice environment, further complicating their use in market research.

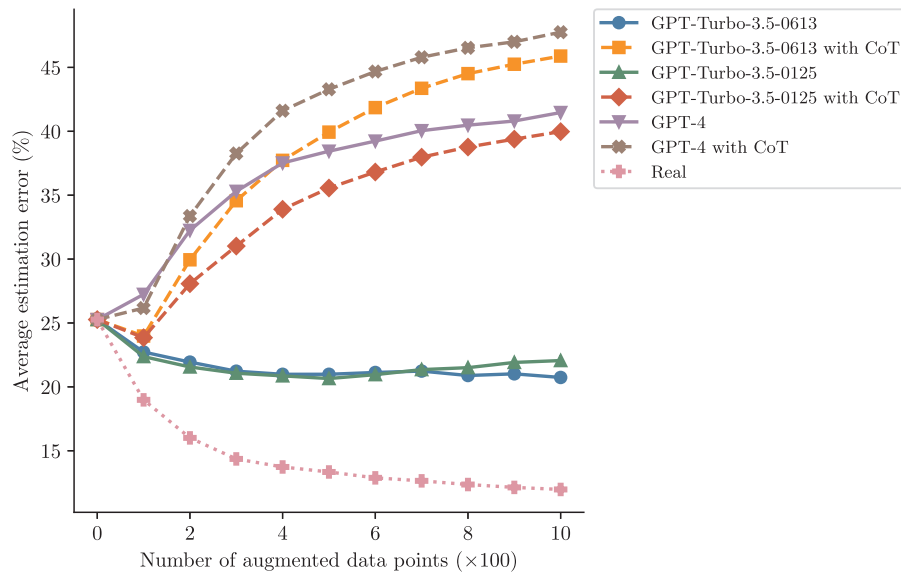
We acknowledge the significant potential of LLM-generated labels for market research, but we also believe that the aforementioned gap is likely to persist, even with the most advanced future developments in AI. Consider conjoint analysis, which is arguably the most fundamental methodology for understanding consumer preferences and decision-making processes in market research (Green and Srinivasan 1990). Typically, the researcher

seeks to understand how consumers value different features or attributes of a product or service (for details on the setup, please refer to Section 3). At its core, customers aim to maximize their benefits by selecting the optimal products based on their past experiences. However, we emphasize that LLMs do not possess real-life experiences. Additionally, consumer preferences evolve over time due to trends, technological advancements, and economic and cultural shifts (Solomon 2020), changes that LLMs may not accurately capture.

The gap between an LLM's predictions and actual customer preferences can lead to misleading results if a researcher simply replaces real data with AI-generated data or naïvely combines the two—a point underscored in Goli and Singh (2024) through careful empirical studies. This observation aligns with our findings. We conducted an empirical study on the conjoint analysis of COVID-19 vaccine preferences, pooling real data with LLM-generated responses based on the same set of product features and performing the conjoint analysis in a manner similar to Brand et al. (2023) (see Section 5 for details of the study). The results are presented in Figure 1. Regardless of the version of the LLM used or whether state-of-the-art prompting techniques are applied, there is a significant gap between the ground-truth model coefficients and those estimated using pooled data. This gap does not diminish with the inclusion of more LLM-generated data; in most models, it actually worsens as more LLM-generated data are added. In contrast, as one would expect, the gap decreases when more real data are incorporated, eventually vanishing to zero, highlighting a stark contrast with the LLM-generated data.

Recognizing that AI-generated data, although informative, is inherently imperfect and likely to be misleading if used naïvely, our goal is to develop a scheme for its efficient utilization. Although engineering techniques such as fine-tuning, CoT prompting, and in-context learning can potentially improve the quality of data from LLMs, our objective is to introduce a statistical approach that can be broadly applied independently of these methods and have theoretical guarantees. Our research question is then as follows: *Can we improve conjoint analysis by effectively integrating LLM-generated data with real data to produce more accurate estimators of customer preferences, compared with using human data, LLM-generated data, or a naïve combination of both?* Our method is inspired by the widely recognized transfer learning and knowledge distillation approach in the deep learning literature (Hinton et al. 2015). By formalizing and adapting their core principles to our context, we present a data-augmentation statistical approach for extracting value from LLM- and AI-generated conjoint data. Specifically, we make two key contributions.

Figure 1. (Color online) LLM-Generated Data $\neq$ Human Data



Notes. The plot illustrates the percentage error of estimation, calculated as $\|(\hat{\beta} - \beta^*)/\beta^*\|_1$, where the ratio is computed component-wise. This error is based on pooling 50 real customer response data points with LLM-generated data of varying sizes. Each curve represents estimators obtained using different combinations of AI-generated or real labels. For further details, please refer to Section 5. The horizontal axis indicates the quantity of data pooled into the model.

1.1. Novel Theoretical Framework for Data Augmentation

We develop a general framework for statistical data augmentation using generative models, such as LLMs, to capture human preferences in conjoint analysis, and propose the AI-augmented estimator (AAE). The framework is broadly applicable to hybrid data sets where true human labels (ground truth) are available for only a subset of data points, whereas AI-generated labels can be created for all observations. These AI-generated labels may be produced by a traditional machine learning (ML) model or a generative model such as an LLM. Although our framework is compatible with any source of surrogate labels, results presented in Section 5 and Online Appendix A show that the estimator performs best when the AI-generated data are more informative. This finding suggests that, at the time of writing, the most effective application of our framework is to augment data generated by LLMs.

Conceptually, our approach leverages the joint distribution among features, surrogate labels, and true labels within the subset of data where both types of labels are observed. This joint structure allows us to learn the distributional discrepancy between surrogate and true labels and subsequently regularize the surrogate labels so that they more closely approximate the true ones. By doing so, we effectively expand the labeled data set, enabling more accurate estimation of human preferences. These principles give rise to theoretically grounded estimators: our estimator is consistent and asymptotically normal (Theorem 4.3) and satisfies a finite-sample, high-probability performance

bound (Theorem 4.6). Furthermore, we demonstrate that our method outperforms approaches that rely solely on human data, solely on LLM-generated data, or on naïve combinations of the two. We also show that our method is a general approach for data augmentation that is broadly applicable: In Section 4.4 and Online Appendix B, we demonstrate that our approach can be extended to the regression setting with a continuous target.

1.2. Large-Scale Empirical Validation

We conduct extensive empirical analyses across multiple conjoint settings to validate the effectiveness and robustness of our proposed AAE. In the first empirical study, based on a real-world COVID-19 vaccination data set, AAE consistently reduces estimation error relative to naïve augmentation and AI-only baselines, even when GPT-generated labels are imperfect. Its regularization capability remains robust across different LLM versions and prompting strategies—basic, CoT, few-shot, and fine-tuned—with GPT-4o and CoT prompting achieving the greatest gains. The results confirm that combining LLM-generated and human data through a statistically grounded augmentation framework yields reliable, unbiased preference estimates while reducing survey data requirements by 24.9%–79.8%. In Section 6, we extend these findings to a separate sports car choice data set and additional robustness checks. AAE again outperforms alternative estimators across all models and prompts, outperforming even fine-tuned GPT-4o baselines. Further comparisons show that AAE matches prediction-powered inference (PPI), an alternative data augmentation strategy (see the literature review and

Angelopoulos et al. (2023a, b)) in low-dimensional cases but surpasses it in high-dimensional conjoint features due to its more granular correction of AI-human discrepancies. Finally, under model misspecification and non-parametric estimation settings, AAE continues to exhibit strong stability and accuracy. Collectively, these empirical results demonstrate that AAE offers a scalable, bias-controlled, and cost-efficient solution for integrating LLM-generated data into real-world market research.

We briefly discuss the organization of our paper. In Section 2, we present a literature review. In Section 3, we set up our model and formally state our data-augmentation approach. We discuss the theoretical properties of our proposed approach in Section 4 and two empirical studies in Sections 5 and 6. In Section 7, we conclude.

2. Literature

In this section, we review several key streams of literature relevant to our research.

2.1. Technical Development of NLP and LLM

The development of LLMs represents a culmination of decades of advances in NLP and deep learning techniques. The foundational work of Chomsky (1956) introduced formal models for language description, paving the way for computational approaches to linguistic analysis. Modern LLMs owe much to significant technical breakthroughs in recent years, such as the introduction of sequence-to-sequence models (Sutskever et al. 2014) and the transformer architecture (Vaswani et al. 2017), which enables efficient handling of sequential data and contextual understanding.

Key milestones in this area include the development of Bidirectional encoder representations from transformers (Devlin et al. 2018), which introduced bidirectional context modeling, and GPT (Radford et al. 2018), which leveraged generative pretraining to advance NLP capabilities. These innovations culminated in LLMs like GPT-4 (Achiam et al. 2023), characterized by their ability to perform complex language tasks after training on massive data sets. Importantly, further improvements, such as domain-specific adaptations as effective strategies for fine-tuning (Beltagy et al. 2019, Gururangan et al. 2020, Parthasarathy et al. 2024) and few-shot learning (Brown et al. 2020) enable these models to achieve remarkable performance in specific tasks. We refer the readers to Naveed et al. (2023) for a review of the broader trajectory of LLM research. These technical advancements underpin the transformative potential of LLMs in applications such as social science and consumer behavior modeling.

2.2. Using LLMs for Social Sciences and Market Research

AI, especially LLMs, is making a pronounced impact on social sciences. Researchers have explored how

these models can simulate human behavior and provide insights into societal phenomena. For instance, Ziems et al. (2024) discussed the potential of LLMs to transform computational social science by generating synthetic data and modeling complex social dynamics. However, questions remain about the reliability of these models in replicating real-world behavior, as highlighted by Huang et al. (2024) and Yang et al. (2024), who cautioned against over-reliance on LLMs for social predictions due to their lack of lived experiences and contextual nuances. Similarly, Gui and Toubia (2023) pointed out challenges in using LLMs to simulate human behavior, especially from a causal inference perspective, where the models may fail to account for underlying causality in decision-making processes.

In the domain of market research, LLMs have emerged as powerful tools for generating synthetic data that mimics human responses. Studies such as Brand et al. (2023) and Argyle et al. (2023) demonstrated the feasibility of using LLMs for simulating consumer preferences, whereas Chen et al. (2023) and Horton (2023) highlighted the potential of these models as simulated economic agents. Still, the extent to which LLMs can faithfully replicate human preferences remains a point of debate, as explored by Goli and Singh (2024), who emphasized the challenges in achieving alignment between AI-generated and human data. Our work provides an alternative perspective on this question: while LLMs may not fully replicate human behavior or replace humans in social sciences or market research, they can nonetheless offer valuable insights to improve existing methods. With the implementation of robust statistical procedures for correcting bias, LLMs can serve as effective tools for enhancing the accuracy and efficiency of current methodologies.

2.3. Conjoint Analysis

Conjoint analysis has long been a cornerstone of market research, providing insights into consumer preferences by evaluating tradeoffs between product attributes (Green and Srinivasan 1978, 1990). Over time, this methodology has seen significant methodological advancements. On the decision-making side, early work by Kohli and Sukumar (1990) and Wang et al. (2009) explored heuristics and optimization approaches for conjoint-based product line design, whereas Kessels et al. (2008) focused on optimizing experimental design for conjoint studies. Recent contributions, such as Dzyabura and Jagabathula (2018), extend the conjoint analysis to multichannel settings, incorporating both online and offline consumer behaviors.

Our research builds on the rich tradition of integrating methods from statistics or ML into conjoint analysis. Although prior studies have leveraged advanced statistical models (Allenby and Rossi 2006) and Bayesian frameworks (Eggers et al. 2022), we introduce a novel data augmentation approach that

bridges the gap between AI-generated and real-world data. Furthermore, as mentioned, Goli and Singh (2024) emphasized the limitations of using LLM-generated data alone for preference estimation, underscoring the need for innovative solutions like our proposed methodology. This methodological contribution represents a significant step forward in enhancing the robustness and scalability of conjoint analysis.

2.4. PPI

Notably, our work also connects to PPI literature. PPI provides a framework for combining a small labeled data set with a large pool of ML predictions to construct unbiased, valid confidence intervals that shrink as prediction quality improves (Angelopoulos et al. 2023b). PPI++ (Angelopoulos et al. 2023a) builds on this by replacing computationally intensive test-based constructions with simple Wald-type intervals, introducing a power-tuning mechanism that automatically adapts to prediction quality and ensuring efficiency gains without risk of degradation when predictions are weak. Most recently, Ji et al. (2025) reinterpret PPI through the surrogate-outcome literature, proposing RePPI, which recalibrates the imputed loss with flexible models to achieve efficiency-optimal inference under convex settings, dominating PPI and PPI++ when recalibration is consistent and still never worse than using labels alone. Together, the three works establish a coherent progression: PPI for validity, PPI++ for computational and adaptive efficiency, and RePPI for optimal use of predictions.

At a high level, AAE and PPI share the same guiding principle: use a small data set containing both real and AI labels to learn the discrepancy between them and then apply this correction to a larger data set containing only AI labels. However, we note two key differences. The key difference lies in how the two methods correct the discrepancy. PPI operates through a rectified score function, subtracting the empirical score function difference from the primary set from the empirical score function of the auxiliary set. However, score function constructed based on log-likelihood functions tend to flatten in high-dimensional feature spaces, which can lead to information loss when adjusting the aggregated loss. AAE, on the other hand, directly models the discrepancy between AI and real data as a function of the feature vector x and AI label z . It corrects AI labels directly by replacing them with predictions of the real labels given the AI label and product features, enabling a more granular adjustment. As shown in our empirical findings in Section 6.3, AAE performs similarly to PPI when x is low dimensional but outperforms PPI in higher-dimensional settings. Please also see Section 4.5 for a detailed comparison between PPI and our method.

2.5. Technology, Innovation, and Entrepreneurship

Our work also speaks to the literature of technology, innovation, and entrepreneurship, the intersection of which has been a focal point of operations management research. Foundational works such as Shane and Ulrich (2004) provide a comprehensive overview of how technological advancements drive product development and entrepreneurial activities. Recent studies emphasize the role of AI and ML in reshaping operational tasks. For instance, Terwiesch (2019) and Bastani et al. (2022) highlight the potential of AI tools to optimize decision-making processes, whereas Yoo et al. (2024) identify new frontiers in digital innovation.

Our work contributes to this stream by demonstrating how LLMs represent a disruptive innovation in market research. By democratizing access to market insights, LLMs enable startups and smaller firms to conduct sophisticated analyses previously reserved for resource-intensive operations (Olsen and Tomlin 2020, Choi et al. 2022). Furthermore, our findings align with Girotra et al. (2023), which discusses the transformative role of LLMs in idea generation and business innovation. Along a similar line, Connell and Choi (2024) discusses how to correct the misclassification error in using LLM in place of human for classifying textual data; Zhang et al. (2023) study debiasing measurement error in ML/AI-generated regressors in partially linear regression models. Ludwig et al. (2024) develops an econometric framework to assess the validity of using LLMs in economics research, emphasizing the need to avoid training data leakage for prediction tasks and to ensure gold standard measurement equivalence for estimation tasks while providing guidance on mitigating potential biases and limitations. Collectively, these insights underscore the potential of new technologies to revolutionize traditional operational practices, particularly for data collection and consumer analysis.

3. Model and the Data-Augmentation Approach

In this section, we establish the theoretical framework. We focus on conjoint analysis as the primary use case to present the theory and conduct the empirical analysis detailed in Section 5.

3.1. Setup

We first provide a detailed explanation of the setup, starting with an in-depth discussion of the theoretical framework underlying the data generation process. We will outline the assumptions, structure, and specific parameters that guide this process, building a foundation for understanding how the data are conceptualized and modeled within this framework.

3.1.1. Data-Generation Process and the Best-in-Class Estimation.

Consider a setting where we observe two data sets. The first, referred to as the *primary data*, consists of m data points, $\{(\mathbf{x}_i^P, y_i^P, z_i^P)\}_{i=1}^m$. For a fixed positive integer k , we denote the set $\{1, \dots, k\}$ by $\mathcal{K}$ and define $\mathcal{K}^+ = \mathcal{K} \cup \{0\}$. These data points are independently and identically distributed (i.i.d.) according to the same distribution as the random vector $(\mathbf{x}, y, z)$, where $y, z \in \mathcal{K}^+$ and $\mathbf{x} \in \mathbb{R}^{dk}$. To clarify, in the context of conjoint analysis with AI-generated labels, $\mathbf{x}$ represents the features or context of a choice setting, and y denotes the respondent's choice. Here, $\mathcal{K}^+$ corresponds to the $k+1$ options available to the respondent, where zero represents the outside option. Additionally, $\mathbf{x} = (\mathbf{x}_{(1)}^\top, \mathbf{x}_{(2)}^\top, \dots, \mathbf{x}_{(k)}^\top)^\top$, where $\mathbf{x}_{(j)}^\top \in \mathbb{R}^d$ corresponds to the features of option j for each $j \in \mathcal{K}$. Throughout this paper, we refer to the random variable y as the *real label*. The variable z represents a predicted label generated by an AI model based on $\mathbf{x}$, and we refer to it as the *AI-generated label*. The second data set, called the *auxiliary data*, consists of $\{(\mathbf{x}_i, z_i)\}_{i=1}^n$, which are independent of the primary data but follow the same distribution. However, in the auxiliary data, the true label y_i is missing for each data point. In Figure 2, we illustrate this theoretical setup using an example simplified from the empirical setting in Section 5. In contrast, in typical conjoint analysis settings, the AI-generated labels z is not available, and our goal is to fit the vector $\boldsymbol{\beta} \in \mathbb{R}^d$ to capture the choice probabilities as accurate as possible using multinomial logit (MNL) models, that is,

$\boldsymbol{\beta}$ such that

$$\sigma_j(\mathbf{x}; \boldsymbol{\beta}) = \frac{e^{\mathbf{x}_{(j)}^\top \boldsymbol{\beta}}}{1 + \sum_{\ell \in \mathcal{K}} e^{\mathbf{x}_{(\ell)}^\top \boldsymbol{\beta}}}, \quad \forall j \in \mathcal{K}, \quad (3.1)$$

best approximates $\mathbb{P}(y|x)$.

How is this data-generation process implemented in practice? A manager facing a conjoint study problem can begin by collecting a small sample of human response data. Using the same questions presented to human respondents, the manager can query an LLM to generate the primary data set containing both labels y and z . Subsequently, the manager can generate additional responses by querying the LLM with a different set of choices, which forms the auxiliary data set where the true label y is missing. Because the cost of querying an LLM is negligible compared with recruiting human subjects in many cases, the auxiliary data are typically much larger than the primary data in size.

We assume that the distribution $\mathbb{P}(y|x, z)$ can be parameterized as $\mathbb{P}(y = j|x, z) = g_j(\mathbf{x}, z; \boldsymbol{\theta}^*)$ for each $j \in \mathcal{K}^+$, where $\boldsymbol{\theta}^* \in \mathbb{R}^q$ is not known to us. This assumption is motivated by the fact that modern artificial AI models, such as an LLM, are trained with vast amounts of text data, including information from numerous sources on the internet, that may contain product reviews, messaging boards, and other online forums with contributions from a wide range of consumers discussing the products they shop for and purchase (Brand et al. 2023). Therefore, AI can “mimic”

Figure 2. (Color online) Illustration of the Data Set

		Efficacy			Protection duration		Major Side-effects		Endorsement				
		50%	70%	90%	1 yr	5 yr	1 in 1M	1 in 10K	President Donald Trump	US Centers for Disease Control and Prevention	Vice President Joe Biden	World Health Organization	
Primary data	$\mathbf{x}_1^P$	$\begin{cases} \mathbf{x}_{1,(1)}^P \\ \mathbf{x}_{1,(2)}^P \end{cases}$	Y		Y		Y		Y				$\begin{cases} z_1^P = 1 \\ y_1^P = 2 \end{cases}$
	$\mathbf{x}_m^P$	$\begin{cases} \mathbf{x}_{m,(1)}^P \\ \mathbf{x}_{m,(2)}^P \end{cases}$	Y			Y	Y				Y	Y	$\begin{cases} z_m^P = 2 \\ y_m^P = 2 \end{cases}$
Auxiliary data	$\mathbf{x}_1$	$\begin{cases} \mathbf{x}_{1,(1)} \\ \mathbf{x}_{1,(2)} \end{cases}$	Y			Y	Y			Y			$\begin{cases} z_1 = 2 \\ y_1 = ? \end{cases}$
	$\mathbf{x}_n$	$\begin{cases} \mathbf{x}_{n,(1)} \\ \mathbf{x}_{n,(2)} \end{cases}$		Y		Y	Y					Y	$\begin{cases} z_n = 1 \\ y_n = ? \end{cases}$

AI: I choose Vaccine #1




Human: Vaccine #2 is better

Notes. In this figure, we illustrate the two data sets, using an example simplified from the empirical setting in Section 5. For each data point in the primary data of size m , two vaccine options are presented to both the human respondent and the AI. Corresponding to the first data point, the first vaccine has the following features: 50% efficacy, a one-year protection duration, a major side effect rate of 1 in 1,000,000, a minor side effect rate of 1 in 10, and endorsement by President Donald Trump. These features can be converted into a numerical feature vector, $\mathbf{x}_{1,(1)}^P$, suitable for conjoint analysis, for example, using one-hot encoding. Note that the options chosen by the human subject and AI may agree with each other (e.g., in the last data point) or differ from each other (e.g., in the first data point). The auxiliary data, which is of size n , is different from the primary data in that we only have the label solicited from the AI.

how respondents make choices, leading to highly informative AI-generated labels. Therefore, the model $(g_j(\mathbf{x}, z; \boldsymbol{\theta}^*) : j \in \mathcal{K}^+)$, which can be chosen as the MNL model or a neural network, is specified to capture the conditional probabilities $\mathbb{P}(y|\mathbf{x}, z)$ in practical scenarios. We also assume throughout that $|g_j(\cdot)| \leq 1$ for all $j \in \mathcal{K}^+$, which is reasonable for a wide range of models.

In this work, we frame the main goal of our conjoint analysis as identifying the *best-in-class* estimator of (3.2)—the one that most closely approximates the true choice probabilities within the set of all MNL models.¹ The quality of this approximation is measured by the Kullback–Leibler (KL) divergence. Formally, the best-in-class estimator is defined as follows:

$$\begin{aligned} \boldsymbol{\beta}^* &\in \arg \min_{\boldsymbol{\beta} \in \mathbb{R}^d} \left\{ \mathbb{E}_{\mathbf{x}} [\text{KL}(\mathbb{P}(y|\mathbf{x}) | \sigma_y(\mathbf{x}, \boldsymbol{\beta}))] \right. \\ &= \mathbb{E}_{\mathbf{x}} \left[\sum_{j \in \mathcal{K}^+} \mathbb{P}(y = j | \mathbf{x}) \log \left(\frac{\mathbb{P}(y = j | \mathbf{x})}{\sigma_j(\mathbf{x}; \boldsymbol{\beta})} \right) \right] \left. \right\}. \end{aligned} \quad (3.2)$$

That is, measuring the loss of approximation using the KL-divergence, $(\sigma_j(\mathbf{x}; \boldsymbol{\beta}^*) : j \in \mathcal{K}^+)$ gives us the approximation to the true choice probabilities that is best in the class of MNL models. Under reasonable assumptions, this minimizer exists and is uniquely defined (see Theorem 4.3).

We further clarify the distinction between our problem and the measurement error issues discussed in the econometrics literature (Bound et al. 2001) and in recent work by Connell and Choi (2024). Specifically, in our context, conditional on $\mathbf{x}$, z should not be interpreted as a noisy realization of y , as is commonly assumed in the measurement error framework. In our setting, $y \perp z | \mathbf{x}$ because the AI cannot access the internal thought and decision-making processes of a human subject, and any relationship between y and z is mediated entirely through $\mathbf{x}$. Nonetheless, we emphasize that our estimation procedure and theoretical results impose no assumptions on the joint distribution of y and z given $\mathbf{x}$. Therefore, even if z indeed represents y measured with error, our approach remains valid.

3.2. Primary, Auxiliary and Naïve Estimators

With no access to auxiliary data such as AI-generated labels, the default approach is to fit the parameter $\boldsymbol{\beta}$ using only the primary data with its real labels $\{(\mathbf{x}_i^P, y_i^P)\}_{i=1}^m$ using the standard maximum likelihood estimation (MLE). We use $\hat{\boldsymbol{\beta}}^P$ to denote the estimator obtained using only the primary data. When the primary data size is sufficiently large, $\hat{\boldsymbol{\beta}}^P$ can be close to $\boldsymbol{\beta}^*$. However, the size of the primary data are often restricted due to the costs of recruiting real subjects. A small primary set yields an *inaccurate* estimator that is far away from $\boldsymbol{\beta}^*$. On the flipping side, we may

directly perform MLE on the auxiliary data $\{(\mathbf{x}_i, z_i)\}_{i=1}^n$, using z_i as labels. Let us define this estimator as $\hat{\boldsymbol{\beta}}^A$. Given the distribution of LLM-generated data and human data are different (i.e., the distributions of y and z , conditioned on $\mathbf{x}$, are different), it is easy to recognize that $\hat{\boldsymbol{\beta}}^A$ could be severely biased regardless of the size of auxiliary data set compared with $\boldsymbol{\beta}^*$.

Because of this limitation, people intend to utilize auxiliary data and AI-generated labels to facilitate the estimation of $\boldsymbol{\beta}^*$. A natural idea for utilizing the AI labels is to simply pool the primary and auxiliary sets. In particular, we define

$$(\tilde{\mathbf{x}}_i, \tilde{y}_i) = \begin{cases} (\mathbf{x}_i^P, y_i^P) & \text{if } 1 \leq i \leq m \\ (\mathbf{x}_{i-m}, z_{i-m}) & \text{if } m+1 \leq i \leq m+n, \end{cases}$$

and perform the standard MLE on $(\tilde{\mathbf{x}}_i, \tilde{y}_i)_{i=1}^{m+n}$. We call this approach the *naïve augmentation* and write the estimator as $\hat{\boldsymbol{\beta}}^{\text{Naive}}$. The auxiliary data largely enlarges the data size. However, because z and y in general come from different distributions, $\hat{\boldsymbol{\beta}}^{\text{Naive}}$ again can be biased, even with large data size. Note that both $\hat{\boldsymbol{\beta}}^P$ and $\hat{\boldsymbol{\beta}}^A$ are both special cases of $\hat{\boldsymbol{\beta}}^{\text{Naive}}$ by setting $n = 0$ or $m = 0$. In the following, we present a simple example to demonstrate the limitations of these estimators discussed above and summarize the observation in Proposition 3.2.

Example 3.1 (AI-Generated Data Cannot Replace Human Data). Consider a simple setting with only one product and no features. Conditional on $z = j$, where $j \in \mathcal{K}^+ = \{0, 1\}$, $y = j$ with probability p . Suppose that $\mathbb{P}(z = 1) = \alpha$. For clarity, we display the joint distribution in Table 1.

Therefore, the probability that $y = 1$ is given by $c_1(\alpha, p) := \alpha p + (1 - \alpha)(1 - p)$ and clearly the best-in-class estimator in (3.1) is given by $\beta^* = \log\left(\frac{c_1(\alpha, p)}{1 - c_1(\alpha, p)}\right)$, which is computed through $\frac{e^{\beta^*}}{1 + e^{\beta^*}} = \mathbb{P}(y = 1) = c_1(\alpha, p)$. Suppose that $n, m \rightarrow \infty$ and $n/m \rightarrow \rho \neq 0$. Asymptotically, the naïve estimator satisfies

$$c_2(\alpha, p) := \frac{\alpha p}{1 + \rho} + \frac{\alpha p + (1 - \alpha)(1 - p)}{1 + \rho} \approx \frac{e^{\hat{\beta}^{\text{Naive}}}}{1 + e^{\hat{\beta}^{\text{Naive}}}},$$

or equivalently $\hat{\beta}^{\text{Naive}} \rightarrow \log\left(\frac{c_2(\alpha, p)}{1 - c_2(\alpha, p)}\right)$. One can verify that unless $\alpha = 1/2$ or $p = 1$, this limit is not equal to β^* . This is always the case even when $n, m \rightarrow \infty$, as long as $\rho \neq 0$. Also note that even when p is close to,

Table 1. Joint Distribution of y and z in Example 3.1

Value of z	$y = 1$	$y = 0$
$z = 1$	αp	$\alpha(1 - p)$
$z = 0$	$(1 - \alpha)p$	$(1 - \alpha)(1 - p)$

but different from, one, z is very informative of y , but the naïve methods still cannot lead to desired estimates.

Proposition 3.2 (Bias of $\hat{\beta}^A$ and $\hat{\beta}^{\text{Naive}}$). *When $m, n \rightarrow \infty$, in general, $\hat{\beta}^A$ and $\hat{\beta}^{\text{Naive}}$ are not consistent estimators of β^* .*

3.3. Estimation with AI-Augmented Data

These challenges lead to our research question: *How do we extract value from the AI-augmented data to facilitate the estimation of β^* ?* To answer this question, we propose the following data augmentation approach that allows us to use the AI-generated data to fit the model.²

3.3.1. Estimation with AI-Augmented Data. Step 1. Obtain an estimator $\hat{\theta}$ to θ^* , where $\mathbb{P}(y = j | \mathbf{x}, z) = g_j(\mathbf{x}, z; \theta^*)$, using the primary data.

Step 2. With the auxiliary data, we construct the estimator $\hat{\beta}^{\text{AAE}}$ as

$$\hat{\beta}^{\text{AAE}} \in \arg \max_{\beta \in \mathbb{R}^d} \left\{ \hat{Q}(\hat{\theta}; \beta) = \frac{1}{n} \sum_{i=1}^n \sum_{j \in K^*} g_j(\mathbf{x}_i, z_i; \hat{\theta}) \log \sigma_j(\mathbf{x}; \beta) \right\}. \quad (3.3)$$

We refer to $\hat{\beta}^{\text{AAE}}$ as the AAE. In the next section, we demonstrate that our estimator $\hat{\beta}^{\text{AAE}}$ successfully recovers β^* and exhibits the desired asymptotic properties. The key idea in our approach is to use the primary data—which contains both LLM-generated and human-produced data—to learn an efficient mapping between human and LLM-generated data. We then leverage this mapping to construct the estimator. This strategy closely resembles a transfer learning approach, where knowledge from one domain (in our case, the LLM-generated data) is transferred to another domain (the human-produced data). The fundamental assumption underpinning such transfer learning techniques is that learning the mapping function between two domains, denoted by $g(\cdot)$ in our context, is inherently simpler and requires fewer data points than carrying out the individual learning tasks within either domain on its own. In other words, in our context, the AI model undertakes the “heavy lifting” of capturing human choice behavior in a fuzzing way, and the first stage in our estimation is trying to capture how fuzzy this AI model is compared with the human model. Moreover, the mapping will become easier to learn with lower LLM costs and higher LLM quality. This is because, as the accuracy and the size of AI-generated labels improves, we anticipate a high-quality estimation of θ due to a decrease in the asymptotic variance of θ estimation; see Section 4.2 and Online Appendix A for further discussions.

We highlight the intuition of our method from two alternative angles. First, our approach is motivated by principles from transfer learning and knowledge distillation, although adapted to the context of conjoint analysis. In transfer learning, knowledge from one domain is leveraged to reduce data needs in another (Pan and Yang 2009, Zhuang et al. 2020); similarly, our method transfers information contained in the joint distribution of product features and LLM-generated labels ($g(\mathbf{x}, z; \theta^*)$) to improve inference on human preferences. Likewise, although classical knowledge distillation involves a student model mimicking a teacher (Hinton et al. 2015), our framework echoes this idea by learning the conditional probability of real labels given LLM labels and product features. Although our method does not involve parameter reuse across tasks or a teacher–student architecture in the narrow sense, the underlying principle is the same: using a large, complex source of information (the LLM) to guide a simpler, statistically grounded estimator. Such a principle can be used to integrate AI-generated labels in other settings, for example, a continuous target in a regression problem (see discussions in Section 4.4 and Online Appendix B).

Second, although LLMs encode vast information, their outputs are systematically different from human responses. Importantly, this discrepancy is often easier to learn from small samples than estimating preferences directly. Our method builds on this intuition: We use a small human data set to capture the gap between human and LLM labels and then apply this correction to the much larger LLM-generated data set. In the AAE algorithm, the first step models the discrepancy between AI and human labels conditional on product features $\mathbf{x}$. The second step then adjusts estimation using the predicted human label based on the AI label and features. When the first-stage model is sufficiently accurate, the second-stage estimator reduces variance by leveraging the enlarged AI-labeled data set (Proposition 4.4). Thus, the more precise the first-stage predictions, the greater the bias reduction in the second stage, as formalized in Theorem 4.6.

4. Theoretical Properties of the AAE and Additional Discussions

This section establishes the large-sample and finite-sample properties of our estimator, $\hat{\beta}^{\text{AAE}}$. We first show that it is consistent and asymptotically normal under mild conditions (Theorem 4.3), extending classical extremum-estimator arguments to the two-stage setting with AI-generated data. We then compare its efficiency with existing estimators. Unlike naïve approaches that are generally biased, AAE is asymptotically unbiased and achieves variance reduction. In particular, Proposition 4.4 shows that the variance of AAE weakly—and often strictly—dominates that of

the plug-in estimator $\hat{\beta}^P$, especially with informative auxiliary data. We provide a finite-sample high-probability bound (Theorem 4.6), which highlights the dependence on first-stage estimation error. In the logistic regression case, we derive a new KL-based risk bound (Proposition 4.8). Overall, AAE enjoys both classical asymptotic guarantees and strong finite-sample performance. Finally, in Sections 4.4 and 4.5, we show that our AAE can be extended to regressions settings and provide a detailed comparison between AAE and PPI, respectively.

4.1. Asymptotic Analysis

To deliver the analysis, a set of common regularity assumptions are needed. The first set of assumptions is used in the analysis of the consistency of $\hat{\beta}^{AAE}$.

Assumption 4.1 (Regularity Conditions for Consistency). *The following assumptions hold.*

- (i) *Identification:* We assume that $\mathbf{x} \in \mathcal{X}$ with probability of one, where $\mathcal{X}$ is bounded. Also, $\mathbb{E}_{\mathbf{x}}[\mathbf{x}\mathbf{x}^\top] \succ 0$ and $\mathbb{P}(y|\mathbf{x}) \geq c$ for all $\mathbf{x} \in \mathcal{X}$, $y \in \mathcal{K}$ and some constant $c > 0$.
- (ii) *First-stage consistency:* $\hat{\theta} \xrightarrow{P} \theta^*$.
- (iii) *Smoothness by continuity:* $(g_j(\mathbf{x}, z; \theta) : j \in \mathcal{K}^+)$ is continuous in θ for all $\mathbf{x} \in \mathcal{X}$ and $z \in \mathcal{K}^+$.

Comparing to those assumptions required in classical consistency results, such as theorem 5.1 in Van der Vaart (2000) or theorem 2.1 in Newey and McFadden (1994), Assumption 4.1 presents similar or even weaker regularity conditions that guarantee consistency in a setting that significantly extends that of classical analysis of MLE under the MNL model, by exploiting the concavity in the problem structure. Part i in Assumption 4.1 essentially serves as an identification condition, which should be satisfied in practical scenarios. Part ii states that the first-step estimator needs to be consistent, which usually is satisfied as long as the models $\mathbf{g}(\cdot) := (g_j(\mathbf{x}, z; \theta) : j \in \mathcal{K}^+)$ are appropriately specified and reasonable estimation strategy such as the MLE is adopted. Part iii is a regularity condition on the functional smoothness. This assumption holds for many widely used models, for example, the MNL model or the neural networks.

Next, the second set of assumptions helps establish asymptotic normality.

Assumption 4.2 (Regularity Conditions for Asymptotical Normality). *We assume that the following conditions hold.*

- (i) *First-stage normality:* $\sqrt{m}(\hat{\theta} - \theta^*) \rightsquigarrow N(\mathbf{0}, \Lambda)$.
- (ii) *Stable sampling ratio:* $\frac{n}{m} \rightarrow \rho$, where $0 < \rho < \infty$.
- (iii) *Smoothness by differentiability:* For each $j \in \mathcal{K}$, $g_j(\mathbf{x}, z; \theta)$ is differentiable for all θ , $\mathbf{x} \in \mathcal{X}$ and $z \in \mathcal{K}$, and there exists an open neighborhood $\mathcal{N}$ of θ^* such that $\mathbb{E}_{\mathbf{x}, z}[\sup_{\theta \in \mathcal{N}} \|\nabla_{\theta} g_j(\mathbf{x}, z; \theta)\|_2] < \infty$.

Assumption 4.2 is again mild. The first item states that $\hat{\theta}$ is asymptotically normal, which is expected. The second item maintains that the ratio between the sample sizes of two data sets needs to be reasonable. The last item is a smoothness condition. Furthermore, to simplify the presentation, some definitions are necessary. We first define $\mathbf{A}(\mathbf{x}; \beta) \in \mathbb{R}^{k \times k}$ with

$$\mathbf{A}(\mathbf{x}; \beta) := \sum_{j \in \mathcal{K}} \sigma_j(\mathbf{x}; \beta) \left(\mathbf{x}_{(j)} \mathbf{x}_{(j)}^\top - \sum_{j' \in \mathcal{K}} \sigma_{j'}(\mathbf{x}; \beta) \mathbf{x}_{(j)} \mathbf{x}_{(j')}^\top \right), \quad (4.1)$$

and $\Omega \in \mathbb{R}^{k \times k}$ and $\Gamma \in \mathbb{R}^{k \times q}$ with

$$\begin{aligned} \Omega &:= \mathbb{E}_{\mathbf{x}}[\mathbf{A}(\mathbf{x}; \beta^*)] \text{ and} \\ \Gamma &:= \mathbb{E}_{\mathbf{x}, z} \left[\sum_{j \in \mathcal{K}^+} \left(\mathbb{1}_{\{j \neq 0\}} \mathbf{x}_{(j)} - \sum_{j' \in \mathcal{K}} \sigma_{j'}(\mathbf{x}, \beta^*) \mathbf{x}_{(j')} \right) \nabla_{\theta} g_j(\mathbf{x}, z; \theta^*) \right]. \end{aligned} \quad (4.2)$$

Further, we let

$$\mathbf{J} := \mathbb{E}_{\mathbf{x}, z} \left[\left(\sum_{j \in \mathcal{K}} (g_j(\mathbf{x}, z; \theta^*) - \sigma_j(\mathbf{x}; \beta^*)) \mathbf{x}_{(j)} \right) \left(\sum_{j \in \mathcal{K}} (g_j(\mathbf{x}, z; \theta^*) - \sigma_j(\mathbf{x}; \beta^*)) \mathbf{x}_{(j)} \right)^\top \right] \in \mathbb{R}^{k \times k}.$$

Equipped with these assumptions and definitions, we show the following key result.

Theorem 4.3 (Consistency and Asymptotic Normality of AAE).

- (i) *Under Assumption 4.1, the optimizer β^* defined in (3.2) is unique, and the AAE satisfies $\hat{\beta}^{AAE} \xrightarrow{P} \beta^*$, when $m, n \rightarrow \infty$.*
- (ii) *Under Assumptions 4.1 and 4.2, it holds that*

$$\begin{aligned} \sqrt{n}(\hat{\beta}^{AAE} - \beta^*) &= \Omega^{-1} \left(\frac{1}{n} \sum_{i=1}^n \sum_{j \in \mathcal{K}} (g_j(\mathbf{x}_i, z_i; \theta^*) - \sigma_j(\mathbf{x}_i; \beta^*)) \mathbf{x}_{(j)} \right. \\ &\quad \left. + \sqrt{\frac{n}{m}} \Gamma \times \sqrt{m}(\hat{\theta} - \theta^*) \right) + o_P(1) \\ &\rightsquigarrow N(\mathbf{0}, \Omega^{-1}(\mathbf{J} + \rho \times \Gamma \Lambda \Gamma^\top) \Omega^{-1}). \end{aligned}$$

Although some aspects of our proof of Theorem 4.3 draw on classical analysis of extremum estimators, the consistency argument we present for a misspecified model with two-stage estimators appears to be novel. First, the identification of the model parameter and its justification (Lemma D.1) is, to the best of our knowledge, new. Second, with the presence of auxiliary data z , we reformulate the loss function as shown in (3.2). This leads to a novel approach that converts the consistency of $\hat{\theta}$ into the uniform convergence of $\hat{Q}(\hat{\theta}; \beta)$ (Lemma D.2). Third, our concavity-based consistency analysis also seems fresh.

4.2. Value of AAE

We compare our proposed estimator, $\hat{\beta}^{\text{AAE}}$, with the three benchmark approaches $\hat{\beta}^{\text{Naive}}$, $\hat{\beta}^{\text{A}}$, and $\hat{\beta}^{\text{P}}$ that were discussed in Section 3.2. With Theorem 4.3 and Proposition 3.2, the merit of $\hat{\beta}^{\text{AAE}}$ over $\hat{\beta}^{\text{Naive}}$ and $\hat{\beta}^{\text{A}}$ is clear: Although $\hat{\beta}^{\text{AAE}}$ delivers asymptotically unbiased estimation, both $\hat{\beta}^{\text{Naive}}$ and $\hat{\beta}^{\text{A}}$ are in general biased. Thus, the use of $\hat{\beta}^{\text{AAE}}$ enables valid estimation, inference, or subsequent downstream optimization, for example, for product design, whereas the other two estimators lead to inferior decisions. Thus, let us focus on the comparison of $\hat{\beta}^{\text{AAE}}$ and $\hat{\beta}^{\text{P}}$ to demonstrate that the convergence speed of $\hat{\beta}^{\text{AAE}}$ being faster than that of $\hat{\beta}^{\text{P}}$.

In this case, standard analysis shows that, under appropriate regularity conditions, the resulting estimator $\hat{\beta}^{\text{P}}$ satisfies $\sqrt{m}(\hat{\beta}^{\text{P}} - \beta^*) \rightsquigarrow N(0, \Omega^{-1} \check{\mathbf{J}} \Omega^{-1})$, where Ω is defined in (4.2), and

$$\check{\mathbf{J}} := \mathbb{E}_{\mathbf{x}, y, z} \left[\left(\sum_{j \in \mathcal{K}} (\mathbb{1}_{\{y=j\}} - \sigma_j(\mathbf{x}; \beta^*)) \mathbf{x}_{(j)} \right) \left(\sum_{j \in \mathcal{K}} (\mathbb{1}_{\{y=j\}} - \sigma_j(\mathbf{x}; \beta^*)) \mathbf{x}_{(j)} \right)^{\top} \right] \in \mathbb{R}^{k \times k}.$$

By Theorem 4.3, the variance of the AAE, $\hat{\beta}^{\text{AAE}}$, can be expressed as $\Omega^{-1} \mathbf{J} \Omega^{-1} / n + \Omega^{-1} \Gamma \Lambda \Gamma^{\top} \Omega^{-1} / m$. Thus, our AAE performs better if and only if

$$\text{Var}^{\text{P}} := \frac{1}{m} \Omega^{-1} \check{\mathbf{J}} \Omega^{-1} \succeq \text{Var}^{\text{AAE}} := \frac{1}{n} \Omega^{-1} \mathbf{J} \Omega^{-1} + \frac{1}{m} \Gamma \Lambda \Gamma^{\top} \Omega^{-1}, \quad (4.3)$$

or equivalently $\check{\mathbf{J}} / m \succeq \mathbf{J} / n + \Omega^{-1} \Gamma \Lambda \Gamma^{\top} / m$. The next result follows.

Proposition 4.4 (Dominance of Var^{AAE}). *Assume Assumptions 4.1 and 4.2 hold and*

$$\Lambda = \mathbb{E}_{\mathbf{x}, y, z} [\nabla_{\theta} \log g_y(\mathbf{x}, z, \theta^*) \nabla_{\theta} \log g_y(\mathbf{x}, z, \theta^*)^{\top}]^{-1}.$$

- (i) *It holds that $\check{\mathbf{J}} \succeq \Gamma \Lambda \Gamma^{\top}$. Therefore, for any $\delta > 0$ and any m , $\text{Var}^{\text{AAE}} \prec \text{Var}^{\text{P}} + \delta \mathbf{I}$ for all n sufficiently large.*
- (ii) *If $\check{\mathbf{J}} \succ \Gamma \Lambda \Gamma^{\top}$, for any m , $\text{Var}^{\text{AAE}} \prec \text{Var}^{\text{P}}$ for all n sufficiently large.*

The assumption we introduce regarding the form of Λ is quite mild—It holds as long as $\hat{\theta}$ is estimated via MLE and the standard regularity conditions are met (Van der Vaart 2000). The first part of the proposition asserts that the variance of AAE weakly dominates the variance of $\hat{\beta}^{\text{P}}$: As long as the auxiliary data set is sufficiently large, the variance of $\hat{\beta}^{\text{AAE}}$ does not exceed that of $\hat{\beta}^{\text{P}}$, up to an arbitrarily small constant. This conclusion follows directly from $\check{\mathbf{J}} \succeq \Gamma \Lambda \Gamma^{\top}$ because we can let n tend toward infinity in (4.3). The argument hinges on

representing $\check{\mathbf{J}} - \Gamma \Lambda \Gamma^{\top}$ as the variance of the residual from projecting $\sum_{j \in \mathcal{K}} (\mathbb{1}_{\{y=j\}} - \sigma_j(\mathbf{x}; \beta^*)) \mathbf{x}_{(j)}$ onto the linear space spanned by $\nabla_{\theta} \log g_y(\mathbf{x}, z, \theta^*)$. For the second part, if $\check{\mathbf{J}} \succ \Gamma \Lambda \Gamma^{\top}$, the strong dominance $\text{Var}^{\text{AAE}} \prec \text{Var}^{\text{P}}$ for all sufficiently large n follows by a similar reasoning.

Is $\check{\mathbf{J}} \succ \Gamma \Lambda \Gamma^{\top}$ generally true? By interpreting $\check{\mathbf{J}} - \Gamma \Lambda \Gamma^{\top}$ as the variance of the residual from projecting $\sum_{j \in \mathcal{K}} (\mathbb{1}_{\{y=j\}} - \sigma_j(\mathbf{x}; \beta^*)) \mathbf{x}_{(j)}$ onto the space spanned by $\nabla_{\theta} \log g_y(\mathbf{x}, z, \theta^*)$, we observe that this inequality should hold in many cases, although providing a general, primitive sufficient condition seems challenging without a specific form of $g(\cdot)$. To illustrate this further and gain more insights, we refer the readers to Online Appendix A, where we show that under specific parametric forms of $g(\cdot)$, item (ii) in the proposition indeed holds in general, and more informative AI-generated labels leads to more variance reduction and therefore higher estimation accuracy.

Finally, we remark that, although our original asymptotic theory assumed $m \rightarrow \infty$, which is unrealistic, it is well known (e.g., via Berry-Esseen's central limit theorem; Feller 1971) that asymptotic variance formulas often provide accurate approximations even at moderate sample sizes. Thus, we believe our asymptotic results still offer meaningful insights into how the AAE estimator reduces variance and bias relative to standard baselines. Moreover, the asymptotic analysis shows that, although the machine data are biased, the correction yields unbiased estimates, an essential property for any debiasing method. In contrast, naïve approaches, such as simply concatenating two data sets, produce biased estimates even with large sample sizes. Motivated by this, we next present a finite sample analysis that is more realistic.

4.3. Finite Sample Analysis

Given that we typically aim to augment a limited amount of human-labeled data with AI-generated data, we extend our theoretical analysis in this section to derive a finite-sample, high-probability performance bound when human data are finite. Throughout, we assume that $\mathbf{x} = (\mathbf{x}_{(1)}^{\top}, \mathbf{x}_{(2)}^{\top}, \dots, \mathbf{x}_{(k)}^{\top})^{\top}$ is a random vector such that each subvector $\mathbf{x}_{(j)}$ has sub-Gaussian norm $\sigma_{\mathbf{x}_{(j)}}$ and is on a bounded support. Without loss of generality, we assume with probability one that $\|\mathbf{x}_{(j)}\|_{\infty} \leq 1$ for all $j \in \mathcal{K}$. Let $\Sigma = \mathbb{E}[\sum_{j \in \mathcal{K}} \mathbf{x}_{(j)} \mathbf{x}_{(j)}^{\top}]$ denote the population covariance, which we assume is positive definite. We further define $\kappa = \mathbb{E}[\max_{j \in \mathcal{K}} \|\mathbf{x}_{(j)}\|_2^2] \leq \text{trace}(\Sigma)$. Additionally, we assume prior knowledge of an ℓ_1 -norm bound on the true parameter vector, that is, $\|\beta^*\|_1 \leq R$ for some known constant R , and we solve (3.3) under the constraint $\|\beta\|_1 \leq R$. These assumptions

are standard in finite-sample analyses of logistic-type models (Wainwright 2019, Chen et al. 2022, Chardon et al. 2024). Because the pecification of $\mathbf{g}(\cdot)$ can vary depending on the application, the performance of the first-stage estimation may differ accordingly. To address this, we introduce the following critical assumption. Let $\tilde{d}$ denote the dimensionality of the parameter space in $\mathbf{g}(\cdot)$.

Assumption 4.5 (First-Stage Performance). *With probability at least $1 - \delta$, it holds that*

$$\mathbb{E}_{\mathbf{x},z}[\text{KL}(\mathbf{g}(\mathbf{x},z,\boldsymbol{\theta}^*)|\mathbf{g}(\mathbf{x},z,\hat{\boldsymbol{\theta}}))] \preceq \epsilon_1(m,\tilde{d},\delta).$$

Under this assumption, we state our key result as follows.

Theorem 4.6 (AAE Finite Sample Bound). *It holds that with probability at least $1 - 3\delta$,*

$$\|\hat{\boldsymbol{\beta}}^{\text{AAE}} - \boldsymbol{\beta}^*\|_2 \leq \frac{(2\sqrt{2\kappa} + 8)e^R(1 + ke^R)^2\sqrt{\epsilon_1(m,\tilde{d},\delta)}}{\lambda_{\min}(\boldsymbol{\Sigma})},$$

provided that (c_1 and c_2 are absolute constants)

$$n \geq \max \left\{ \frac{d \left(\frac{1}{c_2} \log \left(\frac{k}{\delta} \right) \vee 1 \right)}{\frac{\lambda_{\min}(\boldsymbol{\Sigma})}{2K} \wedge 1}, \frac{2d \left(\sum_{j \in \mathcal{K}} \sigma_{\mathbf{x}_{(j)}} \right)^2}{c_1 \epsilon_1(m,\tilde{d},\delta)} \log \left(\frac{4}{\delta} \right) \right\}.$$

Two comments are in order. First of all, a key insight of the theorem is that the error is critically dependent on the error of estimating the conditional probability, $\mathbb{P}(y|\mathbf{x},z)$, in the first stage. Although the specific error bounds vary depending on the model specification, an order of $\sqrt{\tilde{d}/m}$ appears typical. We illustrate this insight with the following example, which applies to a range of potential applications. Second, we note that there is an exponential dependency on the norm bound R . Seemingly, this may indicate that the performance of our AAE deteriorates when the dimensionality d grows large. As formally established in Chardon et al. (2024), in the classical analysis of logistic-type estimators under general sub-Gaussian assumptions on the covariates, that is, $\mathbf{x}$ in our setting, an exponential dependence on the norm bound of the parameter vector is in fact unavoidable. This phenomenon reflects the intrinsic curvature properties of the logistic loss and is not specific to our estimator. Although stronger distributional assumptions on the feature vector—such as exact Gaussianity—can mitigate or remove this exponential dependence as shown in Chardon et al. (2024), our model operates under the more general sub-Gaussian setting, in which such behavior is expected. However, this exponential dependency might be overly conservative: The necessity of exponential dependence on R is often established via carefully constructed, often worst-case, Bernoulli-type designs. In contrast, in many practical settings—particularly when

features are approximately Gaussian—the empirical performance of logistic-type estimators, including AAE, does not degrade as sharply with dimension. Indeed, in such settings, numerical analyses in Section 6.2.2 indicate that, even when d grows reasonably large, the performance of AAE remains stable.

Example 4.7 (Logistic Regression for First Stage). Let us assume that

$$g_j(\mathbf{x},z;\boldsymbol{\theta}) = \frac{e^{\boldsymbol{\theta}^\top \phi_j(\mathbf{x},z)}}{1 + \sum_{j' \in \mathcal{K}} e^{\boldsymbol{\theta}^\top \phi_{j'}(\mathbf{x},z)}} \quad \forall j \in \mathcal{K}.$$

Here, for every j , $\phi_j(\mathbf{x},z) : \mathbb{R}^{d+1} \mapsto \tilde{d}$ is a feature mapping. For a simplistic instance, we may have $\phi_{(j)}(\mathbf{x},z) = (\mathbf{1}\{z=j\}, \mathbf{x}^\top)^\top$, whereas there are many other possibilities. For notation simplicity, let us write $\tilde{\mathbf{x}} = (\tilde{\mathbf{x}}_{(1)}^\top, \tilde{\mathbf{x}}_{(2)}^\top, \dots, \tilde{\mathbf{x}}_{(k)}^\top)^\top$, where $\tilde{\mathbf{x}}_{(j)} = \phi_j(\mathbf{x},z)$ for all $j \in \mathcal{K}$. We assume it has a distribution such that each subvector $\tilde{\mathbf{x}}_{(j)}^\top$ has a sub-Gaussian norm $\sigma_{\tilde{\mathbf{x}}_{(j)}}$ and $\mathbf{x}_{(j)}^\top$'s infinity norm is bounded by one. Also, we assume that it is known that $\|\boldsymbol{\theta}^*\|_1 \leq \tilde{R}$ for some constant $\tilde{R}$ so estimate $\boldsymbol{\theta}^*$ by solving the maximum likelihood problem with the constraint that $\|\boldsymbol{\theta}^*\|_1 \leq \tilde{R}$. Under these assumptions, we derive the following result on the information-theoretic excess-risk inequality, which generalizes that of classical regression or logistic regression analysis based on a localization argument. To our knowledge, it is new. The hidden constants in the result depends on $R, \tilde{R}, k, \sigma_{\tilde{\mathbf{x}}_{(j)}}, \sigma_{\mathbf{x}_{(j)}}, \boldsymbol{\Sigma}$ and κ . Because in typical applications of our approach, the number of products offered to the customer, k , is small, we treat it as a constant and focus on the dependency of our finite sample bound on m , the sample size of primary data, d and $\tilde{d}$, and the number of parameters.

Proposition 4.8 (Finite Sample KL-Excess-Risk Bound). *Under the setup in Example 4.7, with probability at least $1 - \delta$,*

$$\mathbb{E}_{\mathbf{x},z}[\text{KL}(\mathbf{g}(\mathbf{x},z,\boldsymbol{\theta}^*)|\mathbf{g}(\mathbf{x},z,\hat{\boldsymbol{\theta}}))] \preceq \frac{\tilde{d} \log(m/\delta)}{m}.$$

Thus, if n is sufficiently large, with probability $1 - 3\delta$,

$$\|\hat{\boldsymbol{\beta}}^{\text{AAE}} - \boldsymbol{\beta}^*\|_2 \preceq \sqrt{\frac{\tilde{d} \log(m/\delta)}{m}}.$$

When $\tilde{d} \lesssim d$, our estimator enjoys the finite-sample error bound $\|\hat{\boldsymbol{\beta}}^{\text{AAE}} - \boldsymbol{\beta}^*\|_2 \lesssim \sqrt{\frac{d \log(m/\delta)}{m}}$. One can compare existing finite sample analysis of MNL models (Chen et al. 2022) in the classical setting with ours. We observe that our result does not require any assumptions on the magnitude of m , which is typically required to ensure that approximate identification conditions of the loss function is met to approximately

recover β . Instead, equipped with this strong KL bound from Step 1, Step 2 can simply draw a larger auxiliary sample (take n large) to offset any conditioning issues. Therefore, viewed this way, our two-step procedure offers a practical route to high-quality prediction without the stringent sample size and conditioning requirements usually imposed for parameter estimation. In essence, the AAE simply propagates the first-stage risk, measured by the KL divergence, into the second stage.

4.4. AAE for Regression

We remark that our framework of using AI-augmented data to facilitate the estimation and boost the accuracy extends beyond the setting for conjoint analysis. In this section, we extend our approach to the regression setting. The setting is similar to our discussion in Section 3.1. Specifically, we still have two data sets, that is, the primary data, $\{(\mathbf{x}_i^P, y_i^P, z_i^P)\}_{i=1}^m$, and auxiliary data $\{(\mathbf{x}_i, z_i)\}_{i=1}^n$, with the label y_i missing in the auxiliary data. Here we assume that these data points are i.i.d. generated according to the same distribution as the random vector $(\mathbf{x}, y, z)$, where $y, z \in \mathbb{R}$ and $\mathbf{x} \in \mathbb{R}^d$. Similar to the discussion in Section 3.1, here the variable z represents a predicted label generated by an AI model based on $\mathbf{x}$. With a continuous target variable y , we assume that $\mathbb{E}[y|\mathbf{x}, z] = g(\mathbf{x}, z; \theta^*)$, where $\theta^* \in \mathbb{R}^q$. We again seek a best-in-class estimator as

$$\beta^* \in \arg \min_{\beta \in \mathbb{R}^d} \mathbb{E}[(y - \mathbf{x}^\top \beta)^2] \stackrel{(a)}{=} \mathbb{E}[(g(\mathbf{x}, z; \theta^*) - \mathbf{x}^\top \beta)^2], \quad (4.4)$$

where one can verify (a) by expanding the square $(y - \mathbf{x}^\top \beta)^2$ and using the fact that $y - g(\mathbf{x}, z; \theta^*) \perp \mathbf{x}, z$. Note that we do not assume that the regression model $\mathbf{x}^\top \beta^*$ is correctly specified in the sense that it represents the conditional mean of y given $\mathbf{x}$, which is a strong assumption. Instead, (4.4) views it as the best linear predictor of y , or the projection of y on the space of all linear functions of $\mathbf{x}$, and this is always possible. Our proposed estimator, the *AI-augmented regression estimator* (AARE) works as follows.

4.4.1. AARE. Step 1. Obtain an estimator $\hat{\theta}$ to θ^* , using the primary data.

Step 2. With the auxiliary data, we construct the estimator $\hat{\beta}^{\text{AARE}}$ as

$$\hat{\beta}^{\text{AARE}} \in \arg \min_{\beta \in \mathbb{R}^d} \left\{ \hat{Q}(\hat{\theta}; \beta) = \frac{1}{n} \sum_{i=1}^n (g(\mathbf{x}_i, z_i; \hat{\theta}) - \mathbf{x}_i^\top \beta)^2 \right\}.$$

In the discussion in Online Appendix B, we provide a theoretical analysis of AARE. In short, we demonstrate that it possesses the desired asymptotic properties mirroring the key properties of AAE that have been discussed in this section.

4.5. Comparing with PPI

Following our overview in the literature review comparing PPI (Angelopoulos et al. 2023a, b), which is a natural benchmark in our setting, with our approach, we now provide a more detailed discussion. We first briefly describe PPI, specialized to our setting. Recall that in our MNL setting, the target parameter β^* is characterized by the first-order condition $\mathbb{E}[s(\mathbf{x}, y; \beta)] = \mathbb{E}[\sum_{j \in \mathcal{K}} (\mathbb{1}_{\{y=j\}} - \sigma_j(\mathbf{x}; \beta)) \mathbf{x}_{(j)}] = 0$, where we have defined the score function $s(\mathbf{x}, y; \beta) := \sum_{j \in \mathcal{K}} (\mathbb{1}_{\{y=j\}} - \sigma_j(\mathbf{x}; \beta)) \mathbf{x}_{(j)}$. The key idea of PPI is to form an estimating equation by correcting the bias in the score function induced by z . Specifically, PPI replaces the population score with a rectified version that subtracts the empirical discrepancy between the score evaluated at z and at y on the primary sample. This leads to the following estimating equation:

$$\underbrace{\frac{1}{n} \sum_{i=1}^n s(\mathbf{x}_i, z_i; \beta)}_{\text{scores formed by the auxiliary data set using label } z} - \underbrace{\frac{1}{m} \sum_{i=1}^m (\mathbb{1}_{\{z_i^P=j\}} - \mathbb{1}_{\{y_i^P=j\}}) \mathbf{x}_{i,(j)}^P}_{=\frac{1}{m} \sum_{i=1}^m (s(\mathbf{x}_i^P, z_i^P; \beta) - s(\mathbf{x}_i^P, y_i^P; \beta)): \text{the rectified term as the difference between the score evaluated at } z \text{ and at } y \text{ on the primary sample}} = 0.$$

PPI performs inference based on this estimating equation. Equivalently, PPI can be viewed as minimizing a rectified log-likelihood that adds the auxiliary likelihood evaluated at z and subtracts its counterpart on the primary sample. Although this loss-level correction guarantees validity under minimal assumptions, it operates through aggregated score adjustments. As we discuss next, this contrasts with our approach, which directly corrects AI labels conditional on $(\mathbf{x}, z)$ and enables more granular, feature-dependent adjustments.

Our method directly learns the transformation of outcomes conditional on features (including the AI-generated outcome), whereas PPI focuses on subtracting the score function constructed from the loss function of one outcome from another. The key distinction lies in the amount of information retained: by relying only on the score function rather than the full mapping between features and outcomes, PPI risks discarding important information. When no such information is lost, the two approaches may perform similarly—typically in cases with few features and when the loss transformation approximates the outcome transformation closely. However, this equivalence rarely holds in practice. In many realistic settings with high-dimensional features, the PPI loss no longer captures the full mapping, and we find that our method outperforms PPI both in our settings (Sections 5.4 and 6.2), and in theirs, with the performance gap widening as feature dimension increases (Section 6.3).

Moreover, we acknowledge that our ability to capture the full mapping relies on a strong first-stage assumption, which parametrizes the distribution of

outcomes conditional on features and AI labels. Although this assumption underpins our theoretical analysis, our empirical studies show that it does not restrict the practicality of our approach and, in fact, enables AAE to outperform PPI. Moreover, this assumption provides additional flexibility and guarantees a provable performance boost. Importantly, in our framework, z_i need not be the literal “label”—as long as it carries useful information and the parametric assumption holds (In the extreme, z_i could even equal $-y_i$), our method remains applicable. By contrast, the utility of PPI, as emphasized in Angelopoulos et al. (2023b), depends critically on the accuracy of the AI-generated label. Furthermore, Proposition 4.4 establishes the dominance of our estimator over β^P , a discussion absent from the PPI framework.

Finally, the formats of asymptotic and finite sample guarantees of the two approaches are comparable (see Theorems 4.3 and 4.6 in our paper versus theorem 1 in Angelopoulos et al. (2023a) and corollary B.3 in Angelopoulos et al. (2023b)). For example, in finite samples, both methods yield high-probability error bounds that ultimately depend on the informativeness of the AI-generated label. In our setting, this appears as the error term $\epsilon_1(m, \tilde{d}, \delta)$, whereas in PPI, it appears as the width of the rectifier confidence interval ($\mathcal{R}_\alpha^l(\theta)$, $\mathcal{R}_\alpha^u(\theta)$). Although the exact comparison depends on the problem, our empirical results consistently suggest that our approach tends to outperform PPI.

5. Empirical Analysis I: COVID-19 Vaccination

In this section, we present empirical studies to validate our approach. First, we show that a range of LLM models, despite their differences in capabilities, all exhibit substantial misalignment between AI-generated and human-generated data in conjoint analysis. Second, we demonstrate that our proposed method can effectively mitigate this misalignment, leading to improved

performance compared with using only human data or a naïve combination of human and AI-generated data. It is important to emphasize that our method functions as an enhancement to existing models. Even as LLMs continue to improve in their ability to simulate human responses, as long as some degree of misalignment remains between AI-generated and human labels, our approach can further enhance estimation accuracy.

5.1. Empirical Setup

We examine the performance of the AAE based on a high-impact real choice-based conjoint data set for COVID-19 vaccines (Kreps et al. 2020). The study was conducted on July 9, 2020, where 2,000 U.S. adults were recruited to take a 15-minute survey through the Lucid platform. A quota-based sampling was employed to approximate nationally representative samples in terms of demographic characteristics. A total of 1,971 U.S. adults responded to the survey. This survey is important because it provides evidence of factors associated with individual preferences toward COVID-19 vaccination. The results may help inform public health campaigns to address vaccine hesitancy.

The data set consists of responses from 1,971 participants, each expressing preferences for a series of hypothetical vaccines. Every respondent was shown five comparisons between two hypothetical vaccines, described by seven attributes with multiple levels, as outlined in Table 2. Participants were asked to choose one of the two vaccines or opt for neither. We excluded respondents who did not select any vaccines in this setting, as many public LLMs, such as ChatGPT and Gemini, do not permit opting out of vaccines due to safety requirements. Since this data were excluded from both ground truth calculations and LLM data augmentation, this sample selection should not bias our comparisons between estimators.

Data leakage is a critical concern when selecting data sets for LLM-related empirical studies. It occurs

Table 2. Attributes in the Conjoint Analysis

Feature	Levels and description
Efficacy	50%, 70%, and 90% (protection against severe symptoms)
Protection duration	One year and five years
Major side effects	1 in 1,000,000 and 1 in 10,000' (hospitalization or death)
Minor side effects	1 in 10 and 1 in 30 (flu-like symptoms)
FDA approval process	The vaccine has been approved and licensed by the U.S. Food and Drug Administration. The vaccine has received an emergency use authorization from the U.S. Food and Drug Administration. This allows the expedited use of promising drugs that the FDA has found it reasonable to believe may be effective in combatting the virus
National origin of vaccine	China, United Kingdom, and United States
Endorsement	President Donald Trump, U.S. Centers for Disease Control and Prevention, Vice President Joe Biden, and World Health Organization

when an LLM's training data overlaps with the testing data used to evaluate its performance, thereby compromising the validity of the evaluation. In our empirical studies, we use OpenAI's GPT models. Because OpenAI does not disclose its training data, we cannot confirm whether the conjoint data set is part of the GPT models' training data. However, we do not consider this to be a significant issue. If data leakage was present—that is, if the GPT models had already seen the conjoint data set—the naïve augmentation method would perform substantially better. Nevertheless, as we demonstrate later, the naïve augmentation method still produces significant biases, which our method effectively corrects. This highlights the critical importance of our approach.

Using standard MLE, we estimated the best-in-class parameters β^* , as shown in (3.2). For each experiment, we randomly selected 240 respondents from the training set, yielding a data set of 1,200 samples. The vaccine attributes in these samples were converted to text, and various versions of GPT were used to generate labels z_i . Details of the label generation process are provided in Section 5.2. This produced a data set $\mathcal{D}$ with 1,200 data points.

Based on $\mathcal{D}$, we considered different primary and auxiliary data set sizes with $m \in \{50, 100, 150, 200\}$ and $n = 1,000$. For each combination of (m, n) , we randomly sampled $m/5$ respondents from $\mathcal{D}$, using their data as the primary set, $\mathcal{D}^P = \{(x_i^P, y_i^P, z_i^P)\}_{i=1}^m$. From the remaining data set, we sampled $n/5$ respondents and used their vaccine features and GPT-generated labels as the auxiliary set, $\mathcal{D}^A = \{(x_i, z_i)\}_{i=1}^n$. For each GPT model, this resulted in a primary set of size m and an auxiliary set of size n . In the next, we will discuss how to collect the auxiliary data in the conjoint analysis using different GPT models.

5.2. Conjoint Data Generation Using LLMs

In this section, we present our procedure for generating conjoint data using LLMs. We propose a general framework for LLM-based conjoint data generation in Table 3, where the input structure follows three key components to produce conjoint choice data:

This framework allows customization across different prompt designs, conjoint settings, and simulated subject groups, making it adaptable to various LLMs. The structured input ensures consistency in responses. In the subsequent sections, we use this framework to generate choice data with OpenAI's GPT, specifically

GPT-3.5-Turbo-0613, GPT-3.5-Turbo-0125, GPT-4, and the GPT-4o.³ For each model, we generate conjoint data using two prompt engineering approaches: basic prompting and CoT prompting. For GPT-4o, we also implemented the few-shot prompting and fine-tuning.

Before presenting the details of the prompting and fine-tuning techniques in the next, here we remark that our approach is not intended as an alternative to fine-tuning or prompt-based methods for improving LLM outputs. Rather, they are complementary, because our method, AAE, can be applied on top of outputs from models that have already been fine-tuned or employ sophisticated prompt engineering.

5.2.1. Basic Prompting. The basic prompting follows the three-part framework: It starts with an instruction specifying the choice task for GPT. In particular, we ask GPT to act like a random person to simulate the choice from a general population. More detailed demographic information can be added in this part to simulate choices from a more specific population. In the second part of the prompt, we parse the choice task options to text description. We use the minimalist representation of vaccine features, which was shown to be an effective prompting technique (Teixeira 2023). In the last part, we ask which option GPT would choose and require it to return the answer as a single letter. An example of the basic prompt for GPT is shown below in Table 4.

5.2.2. CoT Prompting. Recent advancements in prompting techniques, such as CoT prompting, Tree-of-Thought (ToT), and others, have significantly enhanced reasoning in LLMs. CoT, which structures intermediate reasoning steps, has become the most widely utilized due to its simplicity, interpretability, and robust performance across diverse tasks (Wei et al. 2022). Although methods like ToT, which explores branching reasoning paths (Yao et al. 2023), and other approaches can offer advantages for specific tasks, they often require additional computational resources or fail to consistently outperform CoT across a wide range of domains. Thus, we experiment mainly with CoT for its balance of effectiveness and broad applicability.

We apply this CoT prompting technique with the aim of improving the quality of the choice data generated. Specifically, we introduce conversations that guide GPT to compare the two vaccines feature by

Table 3. LLM-Based Conjoint Data Generation Framework

(I) Choice Task Instruction: Defining the context of the conjoint analysis, specifying the item to choose from, the task for the LLM (selecting from multiple options), and the persona the LLM should emulate.
(II) Choice Task Options Description: Details the attributes of each option in the conjoint choice set.
(III) Choice Extraction: Extracting the choice made by the LLM.

Table 4. Example of the Basic Prompt for GPT

Input (I): You should act like a random person deciding whether to receive a COVID-19 vaccination.
Input (II): There are three options for you:
A: Efficacy: 90%. Protective duration: 5 years. Major side effect: 1 in 10,000. Minor side effect: 1 in 30. Authorization: Approved and licensed by the US Food and Drug Administration. Origin: China. Endorsement: US Centers for Disease Control and Prevention.
B: Efficacy: 50%. Protective duration: 1 year. Major side effect: 1 in 10,000. Minor side effect: 1 in 10. Authorization: Received an emergency use authorization from the US Food and Drug Administration. This allows the expedited use of promising drugs that the FDA has found it reasonable to believe may be effective in combatting the virus. Origin: United Kingdom. Endorsement: World Health Organization. C: You choose neither A nor B.
Input (III): Which option would you choose as a random person? Your response should be a single letter A, B, or C.
Output: A

feature before arriving at a final choice in part (III) of the data-generation framework. An example of the CoT-based conversation is shown below in Table 5.

As the example above illustrates, CoT helps GPT make a more informed choice, and it allows us to know its reasoning process more explicitly. Note that even with CoT, GPT might still make certain mistakes, such as when comparing the magnitude of minor effects in the example above. This type of mistake often happens in arithmetic tasks, and it is hypothesized in the past due to tokenization of these models.

One approach to alleviate this type of situation is to provide further guidance in the reasoning steps, such as indicating “1 in 10” is worse than “1 in 30” regarding minor effects, as shown in the above example. In general, CoT helps GPT to have a more comprehensive evaluation of all choice features, thus becoming more robust against minor reasoning errors.

5.2.3. Few-Shot Prompting. Few-shot prompting is a technique in NLP where a model is guided to perform a specific task by providing a small number of

Table 5. Example of CoT-Based Conversation

Input: Compare the efficacy of A and B, which one is better?
Output: Option A has a better efficacy rate at 90% compared with option B’s 50% efficacy. This means that you would be more likely to be protected against COVID-19 with option A.

Input: Compare the protective duration of A and B, which one is better?
Output: Option A has a protective duration of 5 years, which is longer than Option B’s protective duration of 1 year. Therefore, Option A offers better and longer protection.

Input: Compare the major side effects of A and B, which one is better?
Output: Both Option A and Option B exhibit the same odds of experiencing a major side effect, which is 1 in 10,000. Therefore, neither option is better than the other in this particular aspect. They are equal in terms of the risk of major side effects.

Input: Compare the minor effects of A and B, which one is better? (1 in 30 is lower than 1 in 10)
Output: Actually, Option B is better in terms of minor side effects. The risk of minor side effects for Option B is 1 in 10, which means 1 out of every 10 individuals could experience them. For Option A, it’s higher with 1 in 30 individuals likely to experience minor side effects. In this scenario, a lower ratio (such as 1 in 10) means a higher chance of side effects.

Input: Compare the authorization of A and B, which one is better?
Output: Option A is approved and licensed by the US Food and Drug Administration, which is a more definitive sign of its safety and effectiveness than an emergency use authorization. Option B has received an emergency use authorization, which is a quicker approval process used during public health emergencies, and still shows that the FDA believes it may be effective against the virus but doesn’t represent full approval. Therefore, in terms of authorization, Option A is better as it has full FDA approval.

Input: Compare the endorsement of A and B, which one is better?
Output: Both Option A and Option B have endorsements from highly respected organizations. Option A is endorsed by the US Centers for Disease Control and Prevention, while Option B is endorsed by the World Health Organization. It’s difficult to categorically say which endorsement is better as they’re both credible. However, if you’re based in the US, you might have more trust in the endorsement from the US CDC. Conversely, if you’re based elsewhere in the world or place more trust in a globally recognized body, the WHO’s endorsement might carry more weight for you.

Input: Based on all the previous reasoning, which option would you choose?
Output: Based on efficacy, protective duration, and authorization, Option A seems to be the better choice. Although it has a slightly higher chance of minor side effects, its benefits in terms of providing more robust and longer-lasting protection against COVID-19 and having full FDA approval outweigh the slightly elevated risk. However, each individual’s decision should take into account a multiple set of factors, including their personal health situation and discussions with healthcare professionals.

Input: Return the choice in the previous answer as a single letter A, B, or C.
Output: A

examples directly in the input prompt (Brown et al. 2020). This approach is particularly useful in zero-shot and low-resource scenarios, demonstrating strong performance across a variety of tasks such as text classification, translation, and question answering. We implement the few-shot prompting with GPT-4o by demonstrating 10 survey questions with their true human labels at the beginning of part (I) of the data generation framework. In Table 6, we see an example of the few-shot prompting:

5.2.4. Fine-Tuning. Fine-tuning is another major approach in transfer learning that enhances a model's performance on a specific task with a few examples. In this process, a pretrained model is adapted to a specific task by further training it on new task-specific data. Fine-tuned models have the potential to achieve higher accuracy and better performance for the specific task. In our experiments, we fine-tuned GPT-4o by training it further upon 50 samples of real data.⁴ Basic prompting was employed to generate outputs from the fine-tuned GPT-4o model. Details of the fine-tuning implementation can be found at Peng et al. (2024).

5.3. Estimators and the Evaluation Metric

In this section, we describe the empirical construction and evaluation of our estimators. For each version of the GPT model, we compute the primary-data-only estimator, auxiliary-data-only estimator (i.e., using only AI-generated data), naïve augmentation estimator, and AAE estimator based on $(\mathcal{D}^P, \mathcal{D}^A)$, denoted, respectively, as $\hat{\beta}^P$, $\hat{\beta}^A$, $\hat{\beta}^{\text{Naive}}$, and $\hat{\beta}^{\text{AAE}}$. We also implement the PPI estimator (Angelopoulos et al. 2023b) and the PPI++ estimator (Angelopoulos et al. 2023a), yielding $\hat{\beta}^{\text{PPI}}$ and $\hat{\beta}^{\text{PPI}++}$, respectively. All implementation details and code are provided in <https://github.com/mxw-selene/ai-augmented-estimation>.

Because the other estimators are straightforward to construct, we focus here on the hyperparameter choices for our AAE estimator. In Step 1 of AAE, we

use a small feed-forward neural network to model $(g_j(\mathbf{x}, \mathbf{z}; \boldsymbol{\theta}^*) : j \in \mathcal{K}^+)$, consisting of two hidden layers with ten and five neurons, respectively, and employing the sigmoid activation function for all neurons. We train this network using the Adam optimizer (Kingma and Ba 2014) with a learning rate of 10^{-4} to minimize the cross-entropy loss.⁵ In Step 2 of AAE, we pool both primary and auxiliary data sets to construct the empirical loss function in (3.3) to gain efficiency. Our theoretical results remain valid under this modification, although additional care is required to account for the correlation between the primary data and $\hat{\boldsymbol{\theta}}$.

To assess the estimators, we calculated the MAPE aggregated across all features:

$$\text{MAPE}^l = \frac{1}{d} \left(\sum_{j=1}^d \frac{|\hat{\beta}_j^l - \beta_j^*|}{|\beta_j^*|} \right) \times 100\%,$$

$$\forall l \in \{P, A, \text{Naive}, \text{AAE}, \text{PPI}, \text{PPI}^+\}.$$

For each combination of data set sizes (m, n) , we conducted 50 independent experimental runs and averaged the MAPE to serve as the final performance indicator. We adjust the MAPE values by adding a small constant to the denominator to eliminate the effect of parameters with very small magnitudes. Because MAPE may place greater weight on features with small ground truth coefficients, we also demonstrate that the results remain robust when evaluated using MSE. Details are provided in Online Appendix C.5. In the subsequent discussion, we use MAPE^P , the MAPE based on the primary data set, as the reference point and report the difference in MAPE between each focal estimator and MAPE^P . This difference quantifies the performance gain achieved by employing a given estimator relative to using only the primary data set. For instance, if $\text{MAPE}^{\text{AAE}} - \text{MAPE}^P = -10\%$, it indicates that the AAE estimator reduces the MAPE by 10% compared with using the primary data set.

Table 6. Example of Few-Shot Learning

Input (I): Demonstrations:

You should act like a random person deciding whether to receive a COVID-19 vaccination.

There are three options for you:

A: Efficacy: 70%. Protective duration: 5 years. Major side effect: 1 in 10,000. Minor side effect: 1 in 30. Authorization: Received an emergency use authorization from the US Food and Drug Administration. This allows the expedited use of promising drugs that the FDA has found it reasonable to believe may be effective in combatting the virus. Origin: United States. Endorsement: World Health Organization.

B: Efficacy: 50%. Protective duration: 5 years. Major side effect: 1 in 1,000,000. Minor side effect: 1 in 30. Authorization: Approved and licensed by the US Food and Drug Administration. Origin: China. Endorsement: US Centers for Disease Control and Prevention.

C: You choose neither A nor B.

Which option would you choose as a random person? Your response should be a single letter A, B, or C.

... (the other nine demonstrations).

You should act like a random person deciding whether to receive a COVID-19 vaccination.

5.4. Our Approach vs. Other Approaches of Using LLM-Generated Data

In this section, we present a detailed discussion of the empirical results comparing our estimators to other methods of utilizing LLM-generated data. Table 7 summarizes MAPE difference between various estimators and the MAPE of using only the primary data sets, $\hat{\beta}^P$, that is, $\text{MAPE}^l - \text{MAPE}^P$ for $l \in \{P, A, \text{Naive}, \text{AAE}, \text{PPI}, \text{PPI}^+\}$. Please note that $\hat{\beta}^P$ is the same across all GPT models, with or without CoT, because it employs primary data only. Therefore, the difference in MAPE represents how much gain in MAPE one can get by employing a focal estimator in comparison with simply using the primary data set. Please also refer to Section 5.1 for details of the error metric. The first column of the table indicates the versions of the GPT model. The second column indicates the prompting technique. The rest of the columns show the difference in the MAPE after augmented $n = 1,000$ GPT-generated samples to a real set with m samples with $\hat{\beta}^A$, $\hat{\beta}^{\text{Naive}}$, $\hat{\beta}^{\text{AAE}}$, $\hat{\beta}^{\text{PPI}}$, and $\hat{\beta}^{\text{PPI}^+}$, respectively. Therefore, a *negative* number indicates *error reduction*, whereas a positive number means the error becomes

larger after adding GPT data. Below, we highlight several key observations from the results.

5.4.1. AI-Only Estimation or Naïve Augmentation Incurs More Error.

Our results indicate that relying solely on AI-generated data (yielding estimator $\hat{\beta}^A$) or naively combining AI-generated and human data (yielding estimator $\hat{\beta}^{\text{Naive}}$) can introduce substantial estimation errors relative to simply using primary data set $\hat{\beta}^P$. As shown in Table 7, although $\hat{\beta}^A$ and $\hat{\beta}^{\text{Naive}}$ based on advanced models with advanced prompting techniques may actually increase estimation errors, these simple estimators can reduce error when using more basic models and prompts, such as GPT-Turbo-3.5-0613 and GPT-Turbo-3.5-0125. This finding highlights an inconsistency between an LLM's intrinsic capability and its effectiveness in generating data that resemble human preferences, an observation consistent with prior research (Goli and Singh 2024).

Interestingly, this inconsistency is counterintuitive: The performance of the AI-only estimator and the naïve augmentation approach does not improve with more advanced GPT models or more sophisticated

Table 7. Bias Reduction from $\hat{\beta}^P$ (%)

Model	Prompt	$m = 50$						$m = 100$					
		$\hat{\beta}^A$	$\hat{\beta}^{\text{Naive}}$	$\hat{\beta}^{\text{PPI}}$	$\hat{\beta}^{\text{PPI}^+}$	$\hat{\beta}^{\text{AAE}}$	$p_{\max}$	$\hat{\beta}^A$	$\hat{\beta}^{\text{Naive}}$	$\hat{\beta}^{\text{PPI}}$	$\hat{\beta}^{\text{PPI}^+}$	$\hat{\beta}^{\text{AAE}}$	$p_{\max}$
GPT-3.5 -Turbo-0613	Basic	-5.45	-10.80	—	—	-13.72	8E-11	1.30	-4.53	—	—	-6.90	4E-08
	CoT	20.18	16.81	—	—	-14.79	1E-15	26.93	20.62	—	—	-8.09	3E-12
GPT-3.5 -Turbo-0125	Basic	-8.91	-9.92	—	—	-13.29	2E-14	-2.16	-3.21	—	—	-6.50	2E-09
	CoT	15.67	10.95	—	—	-14.84	2E-15	22.43	14.71	25.28	5.14	-7.72	4E-11
GPT-4	Basic	14.81	12.39	—	—	-15.70	6E-17	21.56	16.20	—	—	-8.04	3E-12
	CoT	21.77	18.12	—	—	-15.80	2E-16	28.53	22.49	23.29	4.13	-8.30	2E-09
GPT-4o	Basic	15.70	13.05	—	—	-15.55	1E-16	22.46	17.34	23.05	4.93	-8.06	2E-12
	CoT	20.61	16.50	—	—	-15.74	2E-16	27.37	20.49	19.30	4.42	-8.06	3E-12
GPT-4o Fine-tuned	FS	12.46	9.71	—	—	-16.16	9E-18	19.22	14.56	17.83	4.46	-8.26	2E-12
	Basic	4.83	3.18	—	—	-16.66	2E-18	11.59	7.96	22.90	4.90	-9.58	2E-14
Model	Prompt	$m = 150$						$m = 200$					
		$\hat{\beta}^A$	$\hat{\beta}^{\text{Naive}}$	$\hat{\beta}^{\text{PPI}}$	$\hat{\beta}^{\text{PPI}^+}$	$\hat{\beta}^{\text{AAE}}$	$p_{\max}$	$\hat{\beta}^A$	$\hat{\beta}^{\text{Naive}}$	$\hat{\beta}^{\text{PPI}}$	$\hat{\beta}^{\text{PPI}^+}$	$\hat{\beta}^{\text{AAE}}$	$p_{\max}$
GPT-3.5 -Turbo-0613	Basic	6.90	-2.06	—	—	-2.09	0.47	7.56	-1.79	—	—	-0.96	0.979
	CoT	32.53	23.06	19.34	2.93	-3.12	2E-05	33.19	21.10	16.08	2.19	-1.82	4E-04
GPT-3.5 -Turbo-0125	Basic	3.43	0.86	24.15	3.48	-1.78	0.006	4.10	1.60	16.70	1.92	-0.76	0.091
	CoT	28.02	17.72	16.44	3.39	-2.74	2E-04	28.69	16.16	10.94	1.95	-1.81	7E-04
GPT-4	Basic	27.16	20.03	11.61	2.91	-3.12	3E-05	27.82	19.20	10.90	1.94	-2.04	2E-04
	CoT	34.12	26.33	13.86	2.95	-3.37	9E-06	34.79	24.10	10.21	1.96	-2.27	3E-05
GPT-4o	Basic	28.05	20.90	13.93	3.23	-3.07	4E-05	28.72	19.65	11.13	1.92	-1.84	2E-04
	CoT	32.96	23.65	13.28	3.29	-3.31	8E-06	33.63	21.28	10.20	1.99	-2.25	5E-05
GPT-4o Fine-tuned	FS	24.81	18.41	11.68	2.78	-3.44	6E-06	25.48	17.50	11.31	2.18	-2.10	9E-05
	Basic	17.18	12.46	13.25	2.87	-4.81	8E-10	17.85	11.31	11.21	1.93	-3.36	5E-09

Notes. For each estimator under each experimental condition, we perform 50 independent runs and report the average MAPE difference between the focal estimator and $\hat{\beta}^P$. The estimators have access to a real data set with m samples and an auxiliary data set containing $n = 1,000$ GPT-generated samples. The reported average MAPE difference thus represents the increase or decrease in MAPE resulting from incorporating these $n = 1,000$ AI-generated samples, relative to using only the m human-labeled samples. For each experimental condition, we also report the maximum p -value from all pairwise t -tests comparing the MAPE of $\hat{\beta}^{\text{AAE}}$ with those of the other estimators under the column $p_{\max}$. The test shows that $\hat{\beta}^{\text{AAE}}$ outperforms other estimators at the 99% significance level for all instances, except the ones marked with a star. —, cases where the numerical value exceeds 1,000, which primarily arises from singularity issues encountered by PPI-based methods when the primary data set is small, leading to estimates dominated by random initialization.

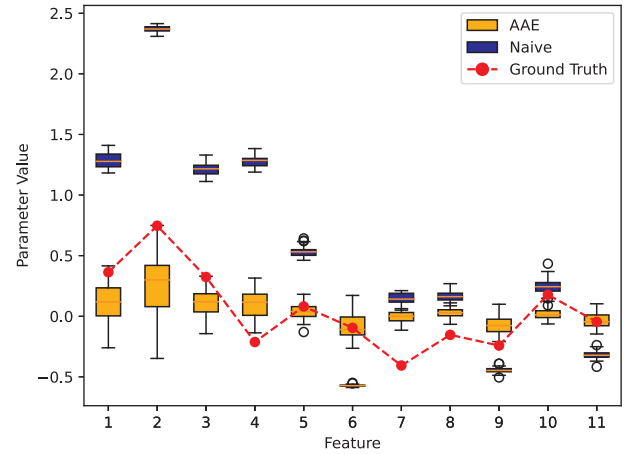
prompting strategies; in many cases, it actually deteriorates. For example, after implementing CoT prompting, we observe that MAPE further increases across models. We conjecture that, in such cases, the LLM becomes “overly rational,” generating choices that may encode richer information but deviate more strongly from human behavior. Although fine-tuning a base model (in this case, GPT-4o) on the primary data set helps mitigate some of the bias introduced by incorporating auxiliary data, it does not fully resolve the issue: including auxiliary data sets still increases MAPE relative to using the primary data set alone.

From Table 7, one can see that the naïve augmentation consistently outperforms pure AI estimation. Indeed, blending in some real data can help align the estimation results. In particular, pure AI estimation can be regarded as a special case of naïve augmentation. In the following, we examine the performance of AAE by comparing it with the naïve augmentation.

5.4.2. AAE Regulates Error Consistently. First, as shown in Table 7, $\hat{\beta}^{\text{AAE}}$ is consistently negative across all models and prompting techniques, indicating that our AAE approach significantly reduces estimation errors relative to using human data alone—regardless of the underlying model or inference quality. We then perform pairwise t -tests comparing $\hat{\beta}^{\text{AAE}}$ with all other estimators in each setting reported in Table 7 and summarize the maximum p -values across these comparisons. The results show that $\hat{\beta}^{\text{AAE}}$ outperforms all benchmark estimators at the 99% significance level in every instance, except for the few cases marked with a star.

More interestingly, unlike the simpler estimators, AAE's performance improves consistently with more advanced versions of GPT models and more sophisticated prompting strategies. It achieves the largest error reduction when applied to the fine-tuned GPT-4o model using all available data. We find that CoT prompting, few-shot prompting, and fine-tuning each contribute to significant error reductions, with fine-tuning being the most effective. As more advanced GPT versions and carefully designed prompts produce more informative AI-generated labels, the effectiveness of AAE correspondingly increases. This observation is consistent with our theoretical discussion in Online Appendix A, which provides formal support for this intuition. This underscores the potential of AAE to deliver superior results with future iterations of LLMs or other AI-based data generators—provided that the augmentation follows a statistically sound estimation strategy such as ours. Indeed, AAE not only regulates the estimation error in AI-generated data but also offers a principled framework for guiding future improvements in AI-assisted estimation.

Figure 3. (Color online) Illustration of $\hat{\beta}^{\text{AAE}}$ and $\hat{\beta}^{\text{Naive}}$ by Feature



Notes. The figures presents the ground-truth parameters β^* (in the dashed curve), $\hat{\beta}^{\text{AAE}}$ (light-colored box) and $\hat{\beta}^{\text{Naive}}$ (dark-colored box). The percentiles in the box plots are based on the 50 experimental runs with $m = 100$.

Furthermore, we observe that AAE generally outperforms the PPI-based methods across different models and prompting strategies. The PPI estimator's MAPE differences are missing for some experimental cases because its performance is unstable due to singularity issues, particularly when the primary data set size m is small. In Section 6.2, we provide a more detailed comparison between AAE and PPI-based methods using both a real data set and a synthetic data set to further investigate the sources of their performance differences.

Finally, beyond the aggregate comparison in Table 7, Figure 3 illustrates the feature-level discrepancies between $\hat{\beta}^{\text{Naive}}$ and $\hat{\beta}^{\text{AAE}}$. The figure compares the estimated parameter values for eleven features across $\hat{\beta}^{\text{Naive}}$, $\hat{\beta}^{\text{AAE}}$, and the ground truth, using $m = 100$ primary data points and $n = 1,000$ auxiliary data points. As shown, the naïve augmentation approach produces substantially larger estimation errors—particularly for the most influential vaccine features—whereas AAE yields estimates that are much more closely aligned with the ground truth.

5.5. Our Approach vs. Human-Only Data—Economic Value of Our Approach

Thus far, we have focused on evaluating our method based on estimation error against other ways of utilizing AI data. In practice, the primary goal of using AI as a data augments in market research is to reduce the costs of hiring real survey participants. Thus, we evaluated the percentage of data saved using AAE with various GPT models, as summarized in Table 8. Specifically, for a given error reduction achieved by applying AAE to a real data set of size n_1 , we

Table 8. Percentage of Saving in Data Size (%)

Model	Prompt	$m = 50$	$m = 100$	$m = 150$	$m = 200$
GPT-3.5-Turbo-0613	Basic	73.59 (0.69)	46.48 (1.64)	30.97 (1.68)	−0.86 (2.80)*
	CoT	76.05 (1.63)	52.84 (2.61)	37.34 (2.93)	8.37 (4.49)
GPT-3.5-Turbo-0125	Basic	71.64 (0.77)	42.67 (1.63)	27.65 (1.36)	−3.72 (2.75)*
	CoT	77.24 (1.23)	52.44 (1.98)	36.22 (2.19)	10.73 (3.77)
GPT-4	Basic	79.30 (1.20)	53.23 (2.28)	39.83 (2.00)	14.01 (3.17)
	CoT	78.40 (1.68)	53.70 (2.89)	40.76 (2.22)	14.96 (3.78)
GPT-4o	Basic	77.86 (1.56)	51.91 (2.74)	38.10 (2.41)	9.39 (4.35)
	CoT	78.81 (1.45)	52.64 (2.66)	41.02 (2.13)	15.14 (3.82)
	FS	79.39 (1.41)	53.79 (2.56)	42.47 (1.97)	13.00 (3.99)
GPT-4o Fine-tuned	Basic	79.81 (1.63)	58.04 (3.22)	50.20 (2.27)	24.86 (4.16)

Notes. This table presents the percentage of data saved using AAE with various GPT models, averaged over 50 experimental runs. The standard errors are shown in parentheses. One sample t -tests show that the savings are all significant at the 95% level except for the ones marked with an asterisk.

calculated the amount of real data samples, n_2 , required to achieve the same error reduction without AI augmentation. The percentage of data saved is then estimated as $((n_2 - n_1)/n_2) \times 100\%$.

Our results show that with fine-tuned GPT-4o, data savings range from 24.9% to 79.8%. When the primary set is small ($m = 50$), AAE saves between 71.64% and 79.8% of data samples, regardless of the GPT model or prompt design. With a moderate primary set size ($m = 100, 150$), savings range from 27.7% to 58.0%, again consistent across GPT models and prompts. Compared with the costs of recruiting real survey participants, the costs of generating AI-based data are negligible and will continue to decrease as generative AI technology advances. In practice, conjoint surveys must be regularly readministered across different product categories and customer segments to account for evolving consumer preferences. This can result in a substantial number of surveys being required over time. Thus, we conclude that AAE offers significant cost savings in the long run.

6. Additional Empirical Analysis

In this section, we present additional empirical studies. In Section 6.1, we assess the robustness of our method on an alternative data set. In Section 6.2, we provide a more detailed comparison between AAE and PPI-based methods. Finally, in Section 6.3, we explore the performance of AAE under model misspecification.

6.1. Empirical Analysis on Other Data

This section presents an alternative empirical study on a real choice-based conjoint data set for sports cars (Spencer 2019) as a robustness check for the results in Section 5. This data set consists of responses from 200 participants, each expressing preferences for a series of hypothetical sports cars. Each respondent was shown 10 sets of three sports cars, described by five attributes with multiple levels, as outlined in Table 9.

Participants were asked to choose one of the three cars. We randomly selected 120 respondents from the training set for each experiment, resulting in a data set of 1,200 samples. The car attributes in these samples were converted to text and used to generate GPT data sets, following a similar procedure to Section 5.2, details of which are discussed in Online Appendix C.2. The remainder of the experimental setup is also consistent with the approach described in Section 5.1.

Table 10 summarizes the change in MAPE of $\hat{\beta}^A$, $\hat{\beta}^{\text{Naive}}$, and $\hat{\beta}^{\text{AAE}}$ in comparison with $\hat{\beta}^P$. Similarly to the findings in Section 5, AAE significantly reduces MAPE in all scenarios, achieving in general the best performance using GPT-4 with CoT and GPT-4o with few-shot prompting, whereas the naïve or AI-only methods tend to elicit errors. Interestingly, AAE's performance shows less correlation with more advanced GPT or prompting techniques compared with the vaccine setting, which is expected given that sports car preferences are highly subjective. Despite being an older version, GPT-3.5-Turbo-0613 performs well with AAE in this context. One can notice that GPT-3.5-Turbo-0613 incurs a small estimation error with naïve augmentation, suggesting it may happen to have a close alignment with real human data, which in turn enhances AAE's effectiveness. Moreover, fine-tuning GPT-4o does not improve the performance of naïve augmentation. This again underscores that fine-tuning is not a universal solution to all problems.

Table 9. Attributes in the Sports Car Conjoint Analysis

Feature	Levels and description
Segment (type of car model)	Basic, fun, and racer
Number of seats	Two, four, and five
Type of transmission	Manual and automatic
Convertible roofs	Yes and no
Price	Price in thousands \$, with three possible choices (30, 35, 40)

Table 10. Bias Reduction from $\hat{\beta}^P$ in Empirical Setting II (%)

Model	Prompt	$m = 50$				$m = 100$			
		$\hat{\beta}^A$	$\hat{\beta}^{\text{Naive}}$	$\hat{\beta}^{\text{AAE}}$	$p_{\max}$	$\hat{\beta}^A$	$\hat{\beta}^{\text{Naive}}$	$\hat{\beta}^{\text{AAE}}$	$p_{\max}$
GPT-3.5-Turbo-0613	Basic	-17.27	-34.20	-37.81	0.105*	15.09	-1.15	-11.02	4e-04
	CoT	148.81	119.98	-34.33	3e-07	181.18	130.32	-9.50	0.002
GPT-3.5-Turbo-0125	Basic	29.70	5.58	-30.22	2e-07	62.06	32.47	-9.80	0.001
	CoT	129.80	112.47	-35.35	9e-08	162.17	124.39	-11.07	6e-04
GPT-4	Basic	-12.43	-16.70	-30.63	2e-04	19.94	10.17	-8.62	2e-04
	CoT	173.34	132.71	-38.32	2e-07	205.71	137.60	-11.36	3e-04
GPT-4o	Basic	421.27	292.59	-25.68	6e-06	453.64	263.41	-8.92	0.006
	CoT	304.98	229.80	-28.65	4e-06	337.34	219.76	-7.86	0.007
GPT-4o Fine-tuned	FS	153.33	89.90	-31.61	2e-07	185.69	98.61	-8.45	0.007
	Basic	201.95	133.85	-36.74	3e-07	234.32	132.10	-12.85	6e-05

Model	Prompt	$m = 150$				$m = 200$			
		$\hat{\beta}^A$	$\hat{\beta}^{\text{Naive}}$	$\hat{\beta}^{\text{AAE}}$	$p_{\max}$	$\hat{\beta}^A$	$\hat{\beta}^{\text{Naive}}$	$\hat{\beta}^{\text{AAE}}$	$p_{\max}$
GPT-3.5-Turbo-0613	Basic	28.53	8.57	-6.83	4e-04	30.71	8.26	-7.09	5e-05
	CoT	194.62	126.11	-4.80	0.019	196.80	119.07	-5.39	5e-04
GPT-3.5-Turbo-0125	Basic	75.50	37.96	-5.30	0.005	77.68	34.69	-6.59	2e-04
	CoT	175.60	123.92	-5.17	0.006	177.78	113.93	-6.64	3e-04
GPT-4	Basic	33.37	21.31	-4.60	0.005	35.55	19.68	-4.72	2e-04
	CoT	219.15	136.34	-7.69	2e-04	221.32	121.56	-6.62	7e-04
GPT-4o	Basic	467.07	234.85	-4.56	0.030	469.25	207.47	-7.03	2e-04
	CoT	350.78	199.59	-5.92	0.010	352.96	178.51	-6.26	1e-04
GPT-4o Fine-tuned	FS	199.13	96.48	-2.84	0.155*	201.31	84.70	-4.38	0.019
	Basic	247.76	123.92	-9.02	2e-04	249.93	110.3	-6.99	4e-04

Notes. For each estimator under each experimental condition, we perform 50 independent runs and report the average MAPE difference between the focal estimator and $\hat{\beta}^P$. In each condition, the estimators have access to a real data set with m samples and an auxiliary data set containing $n = 500$ GPT-generated samples. The reported average MAPE difference thus represents the increase or decrease in MAPE resulting from incorporating these $n = 500$ AI-generated samples, relative to using only the m human-labeled samples. For each experimental condition, we also report the maximum p -value from all pairwise t -tests comparing the MAPE of $\hat{\beta}^{\text{AAE}}$ with those of the other estimators under the column $p_{\max}$. The test shows that $\hat{\beta}^{\text{AAE}}$ outperforms other estimators at the 95% significance level for all instances, except for the ones marked with an asterisk.

Table 11 shows the percentage of data that can be saved using AAE. Our results show data savings ranging from 5.1% to 60.3% across different models, prompts, and primary data sizes. When the primary set is small ($m = 50$), AAE saves between 46.9% and 60.3% of data samples, regardless of the GPT model or prompt design. With a moderate primary set size ($m = 100, 150$), savings range from 18.3% to 40.2%,

again consistent across GPT models and prompts. These results further support the practical value of AAE in broad conjoint settings.

6.2. More Detailed Comparison with PPI-Based Methods

In this section, we present a more detailed comparison between AAE and the PPI-based methods. In addition

Table 11. Percentage of Saving in Data Size in Empirical Setting II (%)

Model	Prompt	$m = 50$	$m = 100$	$m = 150$	$m = 200$
GPT-3.5-Turbo-0613	Basic	59.39 (3.65)	33.26 (5.47)	30.81 (6.53)	17.05 (10.32)*
	CoT	55.04 (4.04)	29.30 (5.50)	26.05 (6.04)	10.47 (10.76)*
GPT-3.5-Turbo-0125	Basic	52.96 (4.31)	29.22 (7.26)	26.29 (6.35)	16.06 (10.71)*
	CoT	56.44 (3.48)	34.18 (4.88)	25.91 (6.99)	14.97 (9.83)*
GPT-4	Basic	48.99 (5.25)	25.74 (7.29)	24.37 (7.20)	5.12 (12.09)*
	CoT	60.29 (3.54)	34.24 (5.26)	34.83 (5.69)	15.81 (8.67)
GPT-4o	Basic	46.90 (4.30)	27.10 (7.07)	23.71 (6.55)	17.20 (9.25)
	CoT	51.39 (3.93)	24.86 (5.88)	29.72 (5.46)	14.20 (9.81)*
GPT-4o Fine-tuned	FS	58.18 (3.42)	38.11 (5.10)	40.23 (5.06)	20.42 (8.60)
	Basic	50.26 (4.93)	27.01 (5.92)	18.34 (6.31)	5.56 (11.90)*

Notes. This table presents the percentage of data saved using AAE with various GPT models, averaged over 50 experimental runs in the second empirical setting. The standard errors are shown in parentheses. One-sample t -tests show that the savings are all significant at the 95% level except for the ones marked with an asterisk.

to the analysis in Section 5 using the COVID-19 vaccine survey data set, we evaluate both approaches on the U.S. private health insurance census data set from the original PPI study (Angelopoulos et al. 2023b). We further investigate their differences using a synthetic data set designed to isolate specific performance characteristics.

6.2.1. Real Data Experiments on Private Healthcare Census Data Set. Table 12 summarizes the performance of AAE and PPI-based methods on the private health insurance census data set used in Angelopoulos et al. (2023b). Because the PPI-based methods suffered from singularity issues under small primary data set in the experiment settings in Section 5, we tested larger primary data size m ranging from 100 to 1,000 in this experiment for a more comprehensive comparison.

Similar to the COVID-19 vaccine setting in Section 5, AAE outperforms PPI-based methods. The performance gap is more pronounced in the vaccine setting. A key difference between the two data sets lies in feature dimensionality: the vaccine data set includes 11 features, whereas the census data set contains only two. As discussed earlier, the distinction between AAE and PPI-based methods is where the correction is applied—at the loss-function level (PPI) versus the mapping-function level (AAE). This implies that performance differences become more significant when the loss function captures less information embedded in the data-generating process, a scenario more likely with higher feature dimensionality. The results from the two data sets are consistent with this reasoning. To further investigate this effect, we conduct a synthetic data study examining how

feature dimensionality impacts the relative performance of AAE and PPI-based methods.

6.2.2. Simulated Data Experiment. We conduct a simulated study to compare AAE and PPI-based methods as the number of features varies. We generated synthetic data sets with varying number of feature d . Details of the synthetic data generation procedure and the experimental setups can be found in Online Appendix C.3. Table 13 summarizes the model performances using these data sets. The results in Table 13 support our previous intuition: when the number of features d is small (i.e., x is low dimensional), the loss function of the outcomes retains much of the information about the mapping function, and the two methods perform similarly. However, as the dimensionality of x increases, AAE significantly outperforms PPI-based methods. These findings are consistent with our results from the previous section.

6.3. Impact of Model Misspecification and Nonparametric Estimates

Because our AAE estimator critically relies on the first-stage estimation of $g(\cdot)$, whose form is usually unclear in practice, we further investigate the impact of model misspecification and nonparametric first stage estimates. To this end, we construct a ground truth model based on a nonparametric tree structure that cannot be well approximated by standard parametric first-stage estimators, thereby inducing model misspecification (see Online Appendix C.4 for details). Using this ground truth model, we evaluate the performance of AAE with different first-stage estimators across various primary set sizes m —We use the logistic regression model and neural networks of different numbers

Table 12. Bias Reduction from $\hat{\beta}^P$ in the Private Healthcare Census Setting (%)

m	$\hat{\beta}^{AAE}$	$\hat{\beta}^{PPI}$	$\hat{\beta}^{PPI+}$	$p_{\max}$
100	−65.86	−22.29	−15.22	0.04
250	−35.98	−12.7	−4.7	0.03
500	−29.93	−7.58	−2.41	0.01
750	−10.74	−2.13	−1.06	0.11*
1,000	−11.87	−4.56	−1.14	0.13*

Notes. For each estimator under each experimental condition, we perform 50 independent runs and report the average MAPE difference between the focal estimator and $\hat{\beta}^P$. In each condition, the estimators have access to a real data set with m samples and an auxiliary data set containing $n = 2,000$ auxiliary samples. The reported average MAPE difference thus represents the increase or decrease in MAPE resulting from incorporating these $n = 2,000$ auxiliary samples, relative to using only the m real samples. Data features are normalized before applying the methods. For each experimental condition, we also report the maximum p -value from all pairwise t -tests comparing the MAPE of $\hat{\beta}^{AAE}$ with those of the other estimators under the column $p_{\max}$. The test shows that $\hat{\beta}^{AAE}$ outperforms other estimators at the 90% significance level for all instances, except for the ones marked with an asterisk.

Table 13. Bias Reduction from $\hat{\beta}^P$ Under Various Dimensionality d (%)

	$\hat{\beta}^{AAE}$	$\hat{\beta}^{PPI}$	$\hat{\beta}^{PPI+}$	$p_{\max}$
$d = 2$	−17.12	−14.01	−17.40	0.53*
$d = 4$	−19.78	−12.35	−16.27	0.06
$d = 6$	−61.41	−29.39	−34.64	3E-08
$d = 8$	−145.71	−5.95	−66.36	2E-08
$d = 10$	−171.98	−80.86	−115.00	5E-06

Notes. For each estimator under each experimental condition, we perform 50 independent runs and report the average MAPE difference between the focal estimator and $\hat{\beta}^P$. In each condition, the estimators have access to a real data set with $m = 100$ samples and an auxiliary data set containing $n = 1,000$ auxiliary samples. The reported average MAPE difference thus represents the increase or decrease in MAPE resulting from incorporating these $n = 1,200$ auxiliary samples, relative to using only the m real samples. For each experimental condition, we also report the maximum p -value from all pairwise t -tests comparing the MAPE of $\hat{\beta}^{AAE}$ with those of the other estimators under the column $p_{\max}$. The test shows that $\hat{\beta}^{AAE}$ outperforms other estimators at the 95% significance level for all instances, except for the ones marked with an asterisk.

Table 14. Bias Reduction of AAE from $\hat{\beta}^P$ Using Different First-Stage Estimators (%)

m	Logistic regression	NN with single hidden layer	NN with two hidden layers
50	−26,612.90	−26,792.41	−26,825.15
100	−386.35	−551.31	−574.82
150	−251.16	−389.32	−406.86
200	−113.72	−236.97	−255.02

Notes. This table reports the bias reduction of AAE with various first-stage estimators, obtained by augmenting m real data points with $n = 1,000$ auxiliary data points. Results are averaged over 50 experimental runs. The large bias reduction at $m = 50$ arises from the nonconvergence of the human-data-only estimator when trained on small data sets.

of hidden layers in our first-stage estimation, capturing varying levels of representational capacity.

As shown in Table 14, estimation error increases with a simpler first-stage model, indicating that model misspecification does affect the empirical performance of AAE. Nevertheless, the overall impact remains moderate compared with using human data only and can be mitigated by employing more flexible first-stage estimators, such as neural networks with increasing complexities, which effectively bring us to a setting with nonparametric first-stage estimates.

Based on our empirical and numerical findings, we offer two practical recommendations. (1) When data are limited, a shallow neural network seem to guard against model misspecification and may serve as an effective choice. (2) When sufficient data are available, one can split the sample and use cross-validation in the first stage to select the optimal model. We also wish to clarify the boundaries of our current study. Our discussion and methodological design remain limited when confronted with challenges such as model misspecification and high dimensionality. In such scenarios, using nonparametric first-stage estimates—such as large and dense neural networks—can be a promising approach. However, in the nonparametric setting, the slower convergence rate—stemming from challenges such as overfitting and the need for regularization—introduces additional practical difficulties. We leave the exploration of these issues to the future research.

7. General Discussion

Looking ahead, our method opens new opportunities for more effective use of LLMs in market research and related domains. As LLM technology continues to advance, this framework provides a foundation for future innovations in data augmentation, helping researchers and practitioners navigate the tradeoffs between AI-generated and human-generated data. We identify several promising directions for future research. First of all, the approach may be more broadly extended to settings where real data are scarce, enabling

new applications of AI for understanding human preferences and behavior. Second, further work could explore additional ways to enhance the practical integration of LLMs with our framework. For example, incorporating different personas into LLM prompts may allow the generated data to more closely reflect the preferences of specific user populations. Third, the proposed framework is not limited to the particular model or data-generation process studied here. Extensions to more general loss functions—beyond the KL divergence or squared loss considered in AARE—represent a natural and important avenue for future research. Fourth, although we note that cross-validation can be used to select the first-stage estimator, alternative or complementary model-selection approaches, such as complexity-based criteria including AIC and BIC, may further improve performance. In principle, if we use out-of-sample predication accuracy as the metric for model selection in the first stage, then this can be exactly in the way as typical ML settings (see Raschka (2018) for a comprehensive discussion). However, the impact of doing so on the second stage is not fully understood, and we call for a systematic investigation of how different model-selection strategies affect empirical outcomes to future work. Finally, as discussed in the concluding section, addressing the slow convergence rates of the first-stage estimator in nonparametric settings could substantially enhance the scalability and applicability of our method, and remains an important direction for future study.

Endnotes

¹ Note that we do not assume that the data are generated by a MNL model. We recognize that choice probabilities parameterized by the MNL model may be mis-specified. In practice, customer choice probabilities can exhibit considerable complexity, yet researchers and practitioners often rely on the simpler MNL model as an approximation in conjoint analyses.

² For simplicity, here we assume that an optimizer $\hat{\beta}^{AAE}$ is obtained. Otherwise, we may optimize approximately and alternatively choose $\hat{\beta}^{AAE}$ such that $\hat{Q}(\hat{\theta}; \hat{\beta}^{AAE}) = \sup_{\beta \in \mathbb{R}^d} \hat{Q}(\hat{\theta}; \beta) - o_P(1)$. In this case, our key theoretical result, Theorem 4.3, is still valid with a slight adaptation of the analysis. Interested readers may refer to the proof of Theorem 4.3 in the Online Appendix.

³ At the time this paper was first written, GPT-5 was not yet available to regular OpenAI API users and was therefore excluded from our empirical analysis. Nevertheless, as discussed earlier, because our method serves as an enhancement to existing models, we do not expect the qualitative conclusions of our results to change as model quality improves—as long as some degree of misalignment between model-generated and human labels remains.

⁴ OpenAI API recommends using 50–100 well-crafted demonstrations, which typically leads to clear improvements.

⁵ We also approximated the $g(\cdot)$ function using alternative methods, including logistic regression and random forests. The qualitative results—namely, that our estimators outperform the alternatives—remained consistent. For brevity, we report only the neural network results.

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